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# The Transmission of Financial Stress to the U.S. Real Economy:

# A BVAR Analysis with Non-Gaussian Identification

Master Thesis in Macroeconometrics

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**Abstract**

Financial crises are tail events that are difficult to predict ex ante, yet capable of generating large and rapid movements in financial markets and the real economy. Due to this disruptive nature, their effects are likely to be particularly visible in high-frequency data. However, a fundamental challenge arises: financial markets respond simultaneously to common information, making the timing and sign restrictions typically used to identify financial shocks difficult to justify.

To address this issue, we identify shocks from the non-Gaussianity of daily innovations, using independent component analysis of the fourth-order cumulant of a daily Bayesian VAR for the S&P 500, the VIX, and the Baa−Aaa credit spread over 1990–2025. This approach requires no zero, sign, or timing restrictions.

Three economically interpretable components emerge as credit, equity, and volatility shocks. Their interpretation is not imposed and is supported by their tail behavior, market responses, crisis timing, and relation to external uncertainty measures. The credit shock is the most non-Gaussian component of the system, despite explaining little of equity-price variance. Aggregating the shocks to monthly frequency reveals distinct macroeconomic transmission: all three are contractionary, but the volatility shock is comparatively short-lived, the credit shock is more persistent and accounts for 21.5% of the twelve-month forecast-error variance of industrial production and 21.5% of prices, while the equity shock operates more gradually through housing.

Finally, we show that the contraction used to reduce the fourth-order cumulant tensor to a matrix can change the ranking of shocks without changing the underlying identified directions, highlighting the importance of invariance in higher-moment identification. Overall, the results show that tail information adds an economically relevant dimension to the covariance-based view of financial shocks. [1][2]

**Keywords:** non-Gaussian identification, independent component analysis, higher moments, temporal aggregation, internal instrument, financial stress, Bayesian VAR.
**JEL:** C32, E44, G01, G10.

*Use of generative AI.* Generative AI tools were used in preparing this thesis for coding support, as a study aid and for English language editing. Details are given in the Statement on the use of generative AI at the end of the thesis.

[1] An online appendix containing additional results, robustness checks, and details is available at https://github.com/leodeco03/Projects-Codes/tree/main/Fin_Shock_ident/Thesis.

[2] All estimation uses the Empirical Macro Toolbox of Ferroni and Canova (2021) (https://github.com/naffe15/BVAR_).

# 1 Introduction

Equity prices, option-implied volatility and corporate credit spreads move together whenever financial conditions deteriorate, and telling apart the disturbances that drive them is a long-standing problem in empirical macroeconomics. The difficulty is simultaneity. The three markets trade in the same session and respond to the same information within the day, so a recursive ordering (Sims, 1980) has no defensible justification at that frequency, and the sign restrictions (Canova and De Nicolò, 2002; Uhlig, 2005) that would separate a credit dislocation from a volatility spike are close to the object one wants to measure. Identification therefore usually proceeds either by constructing a proxy for one shock and leaving the rest unlabelled, or by imposing the response pattern the exercise is meant to uncover.

This paper takes a different route by using daily financial innovations, which often are far from Gaussian: in our sample the pre-whitened innovation to the Baa−Aaa credit spread has an excess kurtosis of 34.3 and a Hill tail index of 2.38. Under the classical result of Comon (1994), mutual independence and non-Gaussianity of the structural shocks point-identify the impact matrix up to permutation and sign, with no restriction on signs, zeros or timing. The tails that a Gaussian likelihood treats as a nuisance are, in this framework, the identifying information, and financial crises are exactly the events those tails record.

Even the sampling frequency is part of our identification strategy. We can interpret temporal aggregation as a smoothing function for kurtosis (our identifying object) that contracts the magnitude that the estimator must consume. For independent innovations, summing $n$ observations divides excess kurtosis by $n$. Volatility clustering slows this decay but does not eliminate it, while aggregation simultaneously shrinks the sample available to resolve an increasingly compressed spectrum.

The economic questions of interest are nevertheless at the monthly frequency, where business cycle dynamics materialize. To bridge the gap between daily and monthly frequencies, we follow the bridge architecture of Gazzani and Vicondoa (2020) and Alessandri et al. (2023): the daily structural shocks are averaged to monthly frequency, which is the filter appropriate to a linear VAR, and injected as *internal* instruments into a monthly macroeconomic BVAR. This estimator identifies relative impulse responses even when the shock is not invertible from the macroeconomic VAR, and classical measurement error in the instrument only scales all responses by a common factor that the normalisation removes (Plagborg-Møller and Wolf, 2021).

The daily identification uses no macroeconomic data, no proxy series and no economic label and it returns three mutually orthogonal statistical objects ordered by non-Gaussianity, making the design agnostic end to end. Every shock then travels to the macro stage, so no shock is selected on a statistic that is later reported as evidence for its importance. Economic meaning is assigned afterwards, from readings that the estimation never saw.

**Results.** Three shocks emerge from the daily system, and four readings that could each have failed separately agree on their labels: own excess kurtosis, the canonical stress signature obtained with no sign restriction, the dates of the largest realisations, and correlations with uncertainty indices that enter no estimation step. The rotation is recovered twice over from 2007–2009 and 2019–2021, two disjoint windows twelve years apart with no observation in

common, and the leading direction agrees to within 17 degrees and anchors on the credit spread in both.

On the second moment the daily system decomposes almost block-diagonally, with one shock owning each market segment. The segment that is quietest in normal times is the one that owns the tails: the credit shock explains 7.0% of equity variance and 0.5% of volatility variance, and carries an excess kurtosis of 37.8, more than three times that of the runner-up. Variance and kurtosis are not two measurements of the same importance, and for a system whose object of interest is financial stress they fall on different shocks.

Once aggregated and brought at monthly frequency all three shocks lower industrial production, housing and prices and raise unemployment, with $[16, 84]$ intervals excluding zero at the peak, although the identification imposes no sign restriction anywhere. Once expressed per common movement of the excess bond premium the troughs are close ($-1.62\%$, $-1.50\%$, $-1.36\%$) and the profiles are not. The volatility shock troughs at three months and recovers, which is the wait-and-see dynamic of Bloom (2009). The credit shock troughs at eleven months, is still visible at four years, and is the only one to contract the *quantity* of credit. The equity shock is slowest and transmits through house prices. The variance decomposition then settles which of them matters: the credit shock accounts for at least 21.5% of the twelve-month forecast error variance of industrial production and 21.5% of that of the price level, against 5.4% and 2.9% for the volatility and equity shocks. The dominance is not uniform, and on unemployment the ordering reverses. The most non-Gaussian component of the daily financial system is therefore also the largest single contributor to medium-run macroeconomic uncertainty, a correspondence that no step in the construction imposes, since the daily stage uses no macroeconomic data and the monthly stage uses no higher moments.

**Contributions.** This thesis makes three:

*(i) Identification at high frequency, transmission at low frequency.* Because aggregation contracts the fourth cumulant, the choice of sampling frequency is a choice about how much identifying information the estimator has. We treat it as such, identify where the non-Gaussianity and the sample size are, and carry the result down in frequency with the filter that preserves it. The two halves are complementary: the high frequency supplies the departure from normality, the higher moments convert it into a rotation, and the bridge converts the rotation into a macroeconomic statement.

*(ii) A fully agnostic identification with a falsifiable protocol for* ex-post *labelling.* No zero, sign or timing restriction is imposed, no real proxy anchors the rotation, and no shock is selected over the others. Because the identification returns unlabelled objects, interpretation becomes a separate inferential act, and Section 4 conducts it on four readings that are logically independent of one another and each falsifiable on its own. Inference is dependence-robust throughout, through a moving-block bootstrap that preserves the volatility clustering the data exhibit, and the paper states when the rotation is identified at all through a near-degeneracy test on the spectrum.

*(iii) The contraction of the cumulant tensor is not innocuous.* Reducing a fourth-order array to a single matrix requires contracting two of its indices against a weight. We show that any

symmetric weight leaves the identified directions untouched and affects only the eigenvalues, hence only the *ordering* of the shocks, which is what every downstream object depends on. The identity weight of Móri et al. (1993) is invariant under a change of pre-whitening basis and returns eigenvalues equal to the shock kurtoses; the all-ones weight of Kollo (2008), standard in applied work, is not invariant, and in these data it would rank the equity shock first and demote the credit shock, reversing a kurtosis ordering of three to one.

**Related literature.** The paper belongs to the strand that identifies SVARs from higher moments. Comon (1994) supplies the statistical result; Lanne et al. (2017) bring it into macroeconometrics through maximum likelihood and, in an extension, allow conditionally heteroskedastic shocks provided their volatility processes are mutually independent; Gouriéroux et al. (2017) develop pseudo-maximum-likelihood inference under i.i.d. shocks, and Lanne et al. (2023) identify from leptokurtosis with a reduced set of moment conditions; Lewis (2025) surveys the field. Drautzburg and Wright (2023) construct inference robust to common stochastic volatility, and we adopt their reading of which co-moment conditions are informative. Montiel Olea et al. (2022) argue that the assumptions this programme imposes on the shock process are substantively strong, and the design here responds to that objection by testing what can be tested, reporting independence diagnostics on the identified shocks rather than assuming them, and by making the labels answer to evidence external to the estimation. Within the higher-moment literature our methodological point concerns the estimator itself: single-matrix contractions of the cumulant tensor are the practical device, the choice of weight is usually treated as a matter of convenience, and we show it decides the ranking.

The second strand is the use of high-frequency information to identify low-frequency systems. Gertler and Karadi (2015) identify monetary shocks from intraday surprises around policy announcements, and Jarociński (2024) shows that the non-Gaussianity of such surprises separates several policy shocks without sign or zero restrictions; Gazzani and Vicondoa (2020) generalise the logic into a bridge procedure that identifies at high frequency, aggregates with the correct filter, and instruments a low-frequency VAR, and Alessandri et al. (2023) apply it to uncertainty shocks. We follow that architecture, with one difference that matters for what the design can claim: the high-frequency stage here is not an event study around a known announcement, so no external surprise series and no event window are required, and the object carried across the bridge is the full set of identified shocks rather than a selected one. The aggregation filter is the one prescribed by Marcellino (1999) and Foroni and Marcellino (2014). At the low-frequency stage we use the internal-instrument estimator of Plagborg-Møller and Wolf (2021) instead of the external proxy-SVAR of Mertens and Ravn (2013), for reasons that are empirical rather than aesthetic: one of our three instruments is partly predictable from lagged real activity, and the internal estimator, which identifies from the component of the instrument that the system's history does not predict, is robust to that failure as well as to non-invertibility.

The third strand is the macroeconomics of financial and uncertainty shocks. Gilchrist and Zakrajšek (2012) identify, through a recursive ordering, a shock to the excess bond premium, the component of corporate spreads not explained by default risk; Jermann and Quadrini (2012) model financial shocks as disturbances to firms' borrowing capacity; Bloom (2009) isolates an uncertainty shock and documents the wait-and-see profile; Caldara et al. (2016) separate

financial from uncertainty shocks with a penalty function and find the financial one dominant; Christiano et al. (2014) attribute close to sixty per cent of the variance of output growth to a risk shock. Our contribution to this debate is that the separation is obtained without sign restrictions, without a penalty function and without a constructed premium, and that it reproduces the received ordering from an identification route that shares no assumption with the existing ones. The estimator also separates an equity component from a volatility component without imposing it. This separation does not map one-to-one onto the decomposition of the VIX into a risk-aversion and an uncertainty part (Bekaert et al., 2013, 2022): in the monthly system the volatility shock accounts for the largest share of both indices, so what the daily stage isolates is a shock to the level of option-implied volatility, distinct from a repricing of equity risk appetite.

Section 2 describes the two panels. Section 3 sets out the identification strategy and the bridge. Section 4 reports the daily results and assigns the economic labels. Section 5 presents the macroeconomic transmission. Section 6 concludes. Derivations, data construction detail, robustness and the full set of diagnostics are in the Online Appendix, whose Section A covers the data, Section B the identification, Section C the bridge, Section D the choice of contraction, Section E the existence of the fourth moment, and Section F additional results.

# 2 Data

The design uses two panels at two frequencies. A daily financial panel serves as the identification stage and a monthly macroeconomic panel as the transmission stage. Online Appendix A records sources, identifiers and every construction detail left out here.

**The daily panel.** Fixed-income series come from the Federal Reserve Economic Data (FRED) database; equity indices come from LSEG Datastream, which is preferred where both providers carry a series because its historical coverage is longer. The baseline system has $M = 3$ endogenous variables,

$$Y_t = \left(\text{SP500}_t,\ \text{VIX}_t,\ \text{CSPREAD}_t\right)', \tag{1}$$

with SP500 and VIX in logarithms and CSPREAD in percentage points. The three span the canonical channels through which financial stress is transmitted and measured: valuation, option-implied uncertainty, and default risk within the corporate segment. Section 3.3 explains why the daily system is kept deliberately small.

The credit variable is the difference between the two Moody's seasoned yield series, $\text{CSPREAD}_t = \text{Baa}_t - \text{Aaa}_t$, rather than the two levels, which would enter as a near-collinear pair driven by a common risk-free factor. The differential isolates compensation for default and downgrade risk net of the term structure. Its distributional shape is what the identification consumes: it sits near a low and stable value through tranquil periods and jumps abruptly during credit dislocations, giving an excess kurtosis of 13.6 in levels (Table 1) and 34.3 once pre-whitened (Table 2), by a wide margin the largest in the system. The Aaa series changes composition on 7 December 2001 and is spliced with a constant level adjustment, which leaves every daily change unaffected and cannot enter a centred cumulant (Online Appendix A.2).

**Why the VIX enters in logarithms.** Volatility is a positive, multiplicatively evolving process whose logarithm is approximately Gaussian (Andersen et al., 2001), so the logarithm is the scale on which its innovations are plausibly symmetric and additive, which is what the linear VAR in (3) assumes. The transformation also decides what the identification selects, and that is the substantive reason for it. The criterion ranks directions by excess kurtosis, and in raw points the size of a move scales with the level, so the largest residuals fall on days when the index was already elevated. Those days are 5 February 2018 and 27 January 2021, the collapse of the inverse-volatility exchange-traded products and the retail short squeeze, on neither of which the Baa−Aaa spread moved appreciably. They are dislocations of the volatility market, not of the economy, unlike October 2008 and March 2020, which moved equity, volatility and credit together. Under the logarithm those two dates remain the largest realisations of the third component, the one with the *smallest* excess-kurtosis eigenvalue, and the leading components are left to the systemically extreme events. With the raw index the ordering inverts outright and the shock carrying the canonical tail-risk signature becomes the least kurtotic component of the system. The numerical record of that comparison, including the conditioning of $\widehat{\Sigma}$ and the whitening check, is in Online Appendix A.3.

**The monthly panel.** The second stage requires a low-frequency system into which the daily shocks can be injected. The panel has $N = 10$ series, with one representative per economic channel:

$$\begin{aligned} X_t = \big(&\mathrm{INDPRO}_t,\ \mathrm{UNRATE}_t,\ \mathrm{HOUST}_t,\ \mathrm{CSHPI}_t,\\ &\mathrm{PCEPI}_t,\ \mathrm{NRCREDIT}_t,\ \mathrm{FFR}_t,\\ &\mathrm{RAI_BEX}_t,\ \mathrm{UNC_BEX}_t,\ \mathrm{EBP}_t\big)', \end{aligned} \tag{2}$$

covering real activity, labour market slack, residential investment and housing wealth, consumer prices, the *quantity* of credit, and the policy rate. The last three series are the excess bond premium of Gilchrist and Zakrajšek (2012), the component of corporate spreads not accounted for by measured default risk, and the risk-aversion and uncertainty indices of Bekaert et al. (2022), which separate the price of risk from its quantity. These three series are ordinary control variables: they are not used in the identification, and they are also the only stationary members of the macro panel, which has consequences for the prior discussed in Online Appendix C.4.

**Alignment, sample and estimation in levels.** The daily series are reindexed onto the NYSE trading calendar and forward-filled on that grid only, so no value is propagated across a weekend or a holiday, giving $T = 9{,}067$ trading days from January 1990 to 2025. The monthly panel covers 432 months over the same span, but estimation stops in September 2025: the federal government shutdown of October 2025 left the unemployment rate permanently unobserved for that month, and the October CPI, an input to the PCE price index, was never collected, and a hole in the middle of a VAR sample silently pairs $y_t$ with the wrong lags. Ending before the gap costs two months and is preferred to interpolating a price index across a month of acute fiscal disruption. The estimation sample is therefore 429 months, 414 usable at $p_m = 15$ lags. Both systems are estimated in levels, without preliminary stationarity transformations, following Sims et al. (1990); for the daily system the argument concerns the conditional mean and not the fourth moments the identification consumes, so it is verified empirically instead

of assumed, and re-running the identification on first differences moves the rotation columns by about one degree (Online Appendix F.3). A separate group of monthly series, the U.S. and global policy-uncertainty indices of Baker et al. (2016) and Davis (2016), and the geopolitical-risk indices of Caldara and Iacoviello (2022), is used only in the external validation of Section 4.5 and enters no estimation step.

**Table 1: Descriptive statistics, daily panel.** Variables in levels, NYSE trading days, $T = 9{,}067$. SP500 and VIX are in logs, CSPREAD in percentage points. The last row reports the raw VIX, which is *not* used in estimation, for the comparison of Online Appendix A.3.

| Variable | Mean | Std | Skew. | Exc. Kurt. |
|---|---|---|---|---|
| SP500 | 7.226 | 0.753 | 0.064 | −0.569 |
| VIX (log) | 2.905 | 0.342 | 0.668 | 0.491 |
| CSPREAD | 1.073 | 0.355 | 2.814 | 13.612 |
| *Not used in estimation* | | | | |
| VIX (raw, points) | 19.456 | 7.785 | 2.200 | 8.630 |

Table 1 already anticipates the analysis. CSPREAD is the only baseline variable with pronounced positive excess kurtosis already in levels; the other two are mildly platykurtic in levels and become strongly leptokurtic once the VAR's linear conditional mean is removed (Section 4.1). And almost all of the raw index's apparent tail thickness is the asymmetry of a bounded-below process, since it disappears under the transformation that makes the process approximately additive.

# 3 Identification strategy

This section sets out what the fourth moments of daily financial data identify, under which assumptions, and how the identified object is carried to monthly frequency. Derivations, the case for each estimation choice against its alternatives, and the implementation detail are in Online Appendix B and Online Appendix C.

## 3.1 The daily VAR and the identification problem

Consider the $M$-dimensional reduced-form VAR of order $p$ for $Y_t \in \mathbb{R}^M$,

$$Y_t = \Phi_1 Y_{t-1} + \cdots + \Phi_p Y_{t-p} + \Phi_0 + u_t, \qquad \mathbb{E}[u_t] = 0, \quad \mathbb{E}[u_t u_t'] = \Sigma_u \succ 0, \tag{3}$$

with $Y_t$ as in (1). Structural shocks relate to reduced-form innovations through the impact matrix,

$$u_t = B^{(0)} \mathbf{v}_t, \qquad \mathbb{E}[\mathbf{v}_t \mathbf{v}_t'] = I_M, \qquad \Sigma_u = B^{(0)} B^{(0)\prime}, \tag{4}$$

and the structural impulse response at horizon $h$ is $\Theta_h = C_h B^{(0)}$, with $C_h$ the moving-average matrices of (3).

The covariance restriction in (4) supplies $M(M+1)/2$ equations for $M^2$ unknowns, leaving $M(M-1)/2$ free parameters: any factorisation $B^{(0)} = \widehat{\Sigma}^{1/2} P$ with $P$ orthogonal satisfies it. Under Gaussianity there is no further information, every orthogonal $P$ is observationally

equivalent, and a recursive scheme resolves the ambiguity only by imposing a causal ordering, which is untenable for variables traded simultaneously at daily frequency. Departures from Gaussianity break the tie.

The daily VAR is estimated under a flat prior, with the lag order selected by marginal likelihood over $p \in \{3, \dots, 10\}$; the criterion returns $p^* = 9$ over a flat surface, and Section 4.1 reads that flatness. The choice of prior is one of minimal commitment, and it is close to immaterial: identification operates through $\Sigma_u$ and the residuals it whitens, and at $T_{\text{eff}} = 9{,}058$ a Minnesota alternative leaves the pre-whitened residuals correlated with their flat-prior counterparts at 0.99998 or above in every equation, with excess kurtoses agreeing to two decimal places. Online Appendix B.1 sets out the argument in full.

## 3.2 Identification from the fourth cumulant

**Assumption 1** (Non-Gaussianity and independence)**.** *The structural shocks* $\mathbf{v}_t = (v_{1,t}, \dots, v_{M,t})'$ *satisfy* (i) $\mathbb{E}[\mathbf{v}_t] = 0$ *and* $0 < \mathrm{Var}(v_{j,t}) < \infty$ *for all* $j$*;* (ii) *the components are mutually independent, not merely uncorrelated, and at most one has a Gaussian marginal;* (iii) *the excess kurtoses* $\kappa_j = \mathrm{cum}_4(v_{j,t})$ *are finite and pairwise distinct.*

**Proposition 1** (Point identification)**.** *Under Assumption 1*(i)–(ii)*,* $B^{(0)}$ *is identified up to permutation and sign of its columns: if* $u_t = B^{(0)}\mathbf{v}_t = \tilde{B}\tilde{\mathbf{v}}_t$ *both satisfy the assumption, then* $\tilde{B} = B^{(0)}\Lambda\Pi$ *with* $\Lambda$ *diagonal non-singular and* $\Pi$ *a permutation (Comon, 1994; Lanne et al., 2017).*

The two residual indeterminacies are resolved by convention and consume no identifying content: the permutation by ordering columns on non-Gaussianity, the sign by the normalisation of Section 3.4. No zero or sign restriction enters at any point. Assumption 1(iii) is the price of extracting $P$ from a single kurtosis matrix instead of jointly diagonalising several, and unlike (ii) it is directly testable on the estimated spectrum.

The object from which $P$ is extracted is the fourth-cumulant tensor of the pre-whitened residuals $\boldsymbol{\iota}_t = \hat{\Sigma}^{-1/2}\hat{u}_t$, $C_{pqij} = \mathrm{cum}(\iota_p, \iota_q, \iota_i, \iota_j)$. Cumulants rather than raw moments, because the subtraction removes the Gaussian part, so that $C = 0$ under normality and whatever remains is identifying signal. Independence has an exact tensor counterpart: writing $\boldsymbol{\iota}_t = W\mathbf{v}_t$ with $W$ orthogonal, mutual independence is equivalent to diagonality of the cumulant tensor in the structural basis, and multilinearity of cumulants then collapses the quadruple sum onto a single index,

$$C_{pqij} \;=\; \sum_{r} \kappa_r \, W_{pr} W_{qr} W_{ir} W_{jr}. \tag{5}$$

The tensor is generated by $M$ numbers, the shock excess kurtoses, together with the rotation. Everything that follows rests on (5).

**From tensor to matrix.** Extracting $P$ with a single eigendecomposition requires reducing a fourth-order array to an $M \times M$ matrix. The standard device is the star product, a linear

contraction of two of the four indices against a symmetric weight $A$,

$$\widehat{M}[p,q] \;=\; \sum_i \sum_j A_{ij}\, C_{pqij} \;\equiv\; (A \star C)[p,q]. \tag{6}$$

Substituting (5) into (6) gives the result that governs the entire strategy.

**Lemma 1** (Any contraction preserves the rotation)**.** *For any symmetric weight $A$, writing $w_r$ for the $r$-th column of $W$,*

$$\widehat{M} \;=\; W \operatorname{diag}\big(\kappa_r\, c_r\big)\, W', \qquad c_r \;=\; w_r' A\, w_r. \tag{7}$$

Because $W$ is orthogonal, (7) is already a spectral decomposition, and two consequences follow at once. The eigenvectors do not depend on the weight, so contracting the tensor costs nothing in the identifying direction. The eigenvalues do depend on it, and only through the scalars $c_r$, so the weight decides the *ordering* of the shocks. Since the ordering is what resolves the permutation indeterminacy of Proposition 1, and every downstream object depends on which shock is called leading, the choice of $A$ is substantive.

We use the identity weight, $A = I$, which gives $c_r = \|w_r\|^2 = 1$ exactly, because the columns of an orthogonal matrix are unit vectors. The eigenvalues of $\widehat{M}$ then *are*, under Assumption 1, the excess kurtoses of the structural shocks. This is the contraction of Móri et al. (1993), equivalently the $K(Z)$ matrix of the FOBI algorithm of Cardoso (1989). The argument for it is an invariance. Consider a change of pre-whitening basis $\boldsymbol{\iota} \mapsto Q\boldsymbol{\iota}$ for orthogonal $Q$, which is what replacing the Cholesky factor by the symmetric square root, or reordering the variables, amounts to. The identity weight is invariant under conjugation, $Q'IQ = I$, so $K(Z) \mapsto QK(Z)Q'$ and its eigenvalues are unchanged. That is as it must be, since the excess kurtosis of a shock is a property of the shock and not of the arbitrary basis chosen to whiten the residuals. A weight failing this invariance makes the labelling of the shocks an artefact of the variable ordering used in the whitening. The all-ones weight of Kollo (2008), the alternative used in applied work, fails it: $Q'\mathbf{1}\mathbf{1}'Q \neq \mathbf{1}\mathbf{1}'$, and its eigenvalues are $\kappa_r s_r^2$ with $s_r = \mathbf{1}'w_r$ the projection of shock $r$ on the direction $(1,\dots,1)$ of the pre-whitened space. Section 4.2 reports what that factor does in these data, and Online Appendix D gives the full comparison.

**Implementation.** The sample counterpart of the fourth-moment matrix of the pre-whitened residuals is

$$\widehat{K} = \frac{1}{T}\sum_{t=1}^{T} (\boldsymbol{\iota}_t \otimes \boldsymbol{\iota}_t)(\boldsymbol{\iota}_t \otimes \boldsymbol{\iota}_t)'. \tag{8}$$

Its excess over the Gaussian counterpart

$$K_z = I_{M^2} + K_{\text{comm}} + \operatorname{vec}(I_M)\operatorname{vec}(I_M)'$$

is contracted with the identity weight, which in this representation amounts to summing the $M$ diagonal blocks of $\widehat{K} - K_z$, and symmetrised. The rotation follows from

$$\widehat{M} = P\,\Lambda\,P', \qquad \lambda_1 \geq \cdots \geq \lambda_M, \tag{9}$$

with columns ordered by descending eigenvalue, so the leading shock is by construction the most non-Gaussian direction. The impact matrix is $B^{(0)} = \widehat{\Sigma}^{1/2}P$ and the shocks are $\mathbf{v}_t = P'\boldsymbol{\iota}_t$. No ridge shrinkage of $\widehat{K}$ is applied, and Online Appendix B.4 shows why none would help: because $K_z$ is invariant under orthogonal conjugation, shrinkage rescales all eigenvalues by a common factor and leaves eigenvectors, ordering, ratios and every downstream object numerically untouched, while biasing the eigenvalues away from the kurtoses they are supposed to estimate.

**The point estimate and the posterior draws.** The chain (8)–(9) is run at two distinct places, and which quantity comes from which matters for how every number below should be read. The *reference identification* is computed once, at the posterior mean of $\Sigma_u$. Under the flat prior that mean is the OLS residual covariance with the degrees-of-freedom correction: the two agree to within 0.21 per cent on every element. The point estimate of $P$, of $B^{(0)}$, of the eigenvalues and of the shock series $\mathbf{v}_t$ is therefore a frequentist ICA-SVAR estimator in the sense of Lanne et al. (2017) and Gouriéroux et al. (2017), implemented through the FOBI eigendecomposition of Cardoso (1989) rather than through their (pseudo-)likelihood, and applied to the OLS residuals. Then every impulse response, variance share, shock realisation and monthly instrument reported in this paper is built from that single rotation.

The 5,000 posterior draws of $\Sigma_u$ produce the bands and never the point. Each draw is passed through the same chain and its rotation $P_s$ is then aligned to the reference one, first by greedy column matching on the absolute inner products $|P_s'P_{\text{ref}}|$, which fixes the permutation, and then by a sign flip whenever $\langle P_{s,k}, P_{\text{ref},k}\rangle < 0$. Both steps are label-fixing operations applied to $P$ and to the corresponding impact column in lockstep, so $B^{(0)}\mathbf{v}_t$ is invariant. Two close eigenvalues swap order under a small perturbation, and the sign of every eigenvector is arbitrary in every draw, so medians taken without a common anchor would average opposite branches towards zero and mix distinct components. Anything reported below as a posterior median with $[16, 84]$ percentiles comes from those aligned draws; anything reported as a bootstrap interval comes instead from the moving-block scheme of Section 3.4, which is the object to trust where the two disagree. Online Appendix B.7 gives the alignment in full.

Lemma 1 supplies a specification check at no cost. Under Assumption 1 the eigenvalues $\lambda_i$ estimate the excess kurtoses $\kappa_i$, so they can be compared with the kurtosis actually realised by the identified shocks. The two are computed from the same data by different routes, one through the whitened residuals and the contraction and one through a univariate moment of the rotated series, and their agreement measures how well the independence that makes (5) collapse holds in the sample. Both are reported in Section 4.1.

## 3.3 Sampling frequency and system dimension

How much identifying information (5) contains is not a fixed property of the three series, depends on the frequency at which they are sampled, which makes the choice of frequency part of the identification design. Write $u^{(n)}$ for the sum of $n$ consecutive innovations, the operation that turns a daily series into a weekly or monthly one. For i.i.d. components, fourth cumulants and

variances are both additive, so

$$\kappa\big(u^{(n)}\big) = \frac{n\,\mathrm{cum}_4(u)}{n^2\mathrm{Var}(u)^2} = \frac{\kappa(u)}{n}, \tag{10}$$

and aggregation is a contraction on precisely the object the estimator consumes. Equation (10) is a benchmark and not a description of financial data: volatility clusters, extreme days arrive together, and under GARCH-type dependence the decline is far slower than $1/n$ (Drost and Nijman, 1993). Aggregating our daily residuals over 21 trading days leaves an excess kurtosis of 6.90 for the S&P 500 and 10.58 for the credit spread, against the 0.48 and 1.65 that independence would predict.

What deteriorates decisively is not the level of the fourth moment but the precision with which the spectrum can be resolved. Identification rests on the separation between adjacent eigenvalues of the kurtosis matrix, and aggregating by $n$ acts on that separation twice, compressing the eigenvalues towards one another and dividing the sample by $n$, so the standard error of each estimate grows as the gap it must resolve shrinks. The same residuals aggregated to quarterly frequency leave 143 observations, at which point the estimated ordering is no longer stable. This, and not the absence of non-Gaussianity, is why higher-moment identification has found its most convincing applications in high-frequency financial data.

The daily system is also deliberately small, and dimension enters this design differently from the way it enters a conventional VAR. The identifying object has $M^4$ entries, 81 at $M = 3$ against 2,401 at $M = 7$, estimated from the same $T$, and the per-entry sampling error of a fourth moment is already large under heavy tails. The binding consequence falls on Assumption 1(iii): a wider daily financial system supplies its additional components from blocks that are close to redundant, since a set of yields and spreads is in substance a low-dimensional factor structure, so the extra components are near-Gaussian, carry small and mutually similar kurtoses, and separate only weakly. Degeneracy at the foot of the spectrum is what a larger system produces. The apparent cost, omitting channels, is not a cost in a two-stage design: the daily VAR is an identification device whose job is to span the directions that generate the tails, and economic breadth is supplied by the ten-variable monthly system of Section 3.5.

## 3.4 What must hold, and how it is checked

**Sign normalisation.** The sign of each column of $B^{(0)}$ is not identified and is fixed by convention, in the original-unit impact space rather than in the whitened space. For each shock $k$ we take the variable whose *standardised* impact response is largest,

$$m^*(k) = \arg\max_m \left|B^{(0)}_{mk}\right| \big/ \sigma_m, \qquad \sigma_m = \left(\textstyle\sum_k B^{(0)2}_{mk}\right)^{1/2}, \tag{11}$$

and fix the sign so that this dominant impact is positive. Standardising matters, because an argmax over raw $|B^{(0)}|$ is dominated by whichever variable has the largest unit scale. This is a normalisation and not a restriction: replacing $(B^{(0)}, \mathbf{v}_t)$ by $(B^{(0)}S, S\mathbf{v}_t)$ with $S = \mathrm{diag}(\pm 1)$ leaves the reduced form, the likelihood and every historical contribution unchanged, so no draw is discarded and no response is constrained. It is nonetheless necessary, because the sign of each

eigenvector is arbitrary in every posterior draw and medians taken without a common anchor would average opposite branches towards zero. The convention fixes the meaning of the *positive* direction of each shock; the direction realised on any given date is a data object.

**Near-degeneracy.** Assumption 1(iii) is the one most likely to fail, and it is testable. Point identification of shock $i$ requires its eigenvalue to be separated from its neighbours, since with $\lambda_i \approx \lambda_{i+1}$ the corresponding eigenvectors are identified only jointly, up to an arbitrary rotation within the degenerate subspace. For each adjacent pair we report the posterior probability of strict ordering $\Pr(\lambda_i > \lambda_{i+1} \mid \mathbf{Y})$, the posterior median and 16th percentile of the gap, and the concentration ratio $\lambda_1/\lambda_2$; a pair is flagged below 0.95. Because the identity weight makes the eigenvalues equal to the kurtoses, these quantities carry a direct economic reading that would not survive a basis-dependent weight: $\Pr(\kappa_{(1)}/\kappa_{(2)} > 2 \mid \mathbf{Y})$ measures how uniquely non-Gaussian the leading shock is, which in a scheme that orders shocks by non-Gaussianity is the relevant notion of identification strength. The literature offers no standard test here, and a spectrum of similar eigenvalues identifies nothing and says so.

**Independence, and which diagnostics are informative.** Assumption 1(ii) is what the estimator imposes, so it cannot be tested directly, but its observable implications can be examined on the identified shocks. Identification is a statement about the contemporaneous distribution of $u_t$ (Comon, 1994), and the sample moment (8) converges to its unconditional counterpart under stationarity and ergodicity, so serial dependence within each shock inflates the sampling variance of $\widehat{K}$ without biasing $P$; Lanne et al. (2017) show that identification survives conditionally heteroskedastic shocks provided their volatility processes are mutually independent. Dependence across shocks through a common volatility factor is different, since it violates Assumption 1(ii), but for the identity contraction it is nonetheless harmless: if $v_t = \omega_t^{1/2} z_t$ with $z_t$ mutually independent, zero-mean and independent of $\omega_t$, every off-diagonal entry of the Móri matrix in the structural basis contains a component of $z_t$ to the first power and vanishes, so the eigenvectors are unchanged. The eigenvalues become $d_i = \kappa_i + \sum_{j \neq i} \left( \mathbb{E}[v_i^2 v_j^2] - 1 \right)$ (Online Appendix B.6).

Contemporaneous independence requires more care. The natural statistic is the co-kurtosis $\mathbb{E}[v_{it}^2 v_{jt}^2]$, which equals one under independence, and it is not robust to common stochastic volatility. Let $v_{i,t} = \sqrt{\omega_t}\, z_{i,t}$ with $z$ conditionally independent and symmetric and $\omega_t > 0$ a common volatility factor. Then

$$\mathbb{E}[v_i^2 v_j^2] = 1 + \mathrm{Var}(\omega) > 1, \qquad \mathbb{E}[v_i^2 v_j] = 0, \qquad \mathbb{E}[v_i^3 v_j] = 0. \tag{12}$$

A positive common factor inflates even powers of the residual component, whereas odd powers are annihilated by the symmetry of $z$ and survive as genuine tests. A rejection on the co-kurtosis is therefore expected under conditionally independent shocks driven by common volatility and carries no evidence against Assumption 1(ii); the informative statistics are the co-skewness and the asymmetric fourth moment, both with null value zero. This is the logic by which the SV-robust GMM of Drautzburg and Wright (2023) discards the co-kurtosis moment condition. We report all three panels and base the assessment on the two odd-power ones.

**Dependence-robust inference.** Posterior draws of $\Sigma_u$ treat $\widehat{K}$, and hence the eigenvalues that order and identify the shocks, as if estimated from i.i.d. data. Honest bands for any object built from fourth moments require a scheme that preserves serial dependence, so inference uses a moving-block bootstrap with block length $\ell = 20$ trading days, about one trading month of volatility persistence. It is applied twice: to the pre-whitened residuals, re-running the chain (8)–(9) on 500 resamples to obtain intervals for each $\lambda_i$, the bootstrap probability of leading-shock dominance, and a column instability index measuring how stable each identified direction is across resamples; and to the identified shock series themselves, with 5,000 replications, to obtain intervals for the odd-power co-moments in (12). The second application carries more weight than its routine appearance suggests, because those statistics have null value zero and an i.i.d. scheme would produce intervals far too narrow and manufacture rejections of an assumption that holds. The two applications separate the two things that can go wrong with an identification of this kind, an ill-determined spectrum and a failure of contemporaneous independence, and price each under the dependence the data exhibit. Online Appendix B.8 gives the implementation.

## 3.5 The bridge to monthly frequency

The daily analysis identifies shocks and characterises their propagation within the financial system over days, such that it cannot speak to output, employment or prices. The second stage aggregates the identified shocks to monthly frequency and injects them into a macroeconomic BVAR as an internal instrument.

There is no selection step anywhere in the pipeline: the identification returns the $M$ columns of the rotation in one go and all of them are carried forward, in their natural order. Selecting the shock to instrument on the basis of a statistic, its own kurtosis or its correlation with a real proxy, and then reporting that same statistic as evidence of the shock's importance, would be circular. Keeping all three is what allows Section 5.3 to report the macroeconomic importance of each without that importance having been built into which shock was carried across.

**The aggregation filter.** A daily shock cannot be used as a monthly instrument without an aggregation rule, and the rule is not innocuous. The bridge literature establishes that the high-frequency object must be aggregated with the filter matching how the low-frequency variable is sampled (Marcellino, 1999). We use the signed monthly mean,

$$Z_m^{(k)} = \frac{1}{D_m} \sum_{t \in m} v_{k,t}, \tag{13}$$

with $D_m$ the number of trading days in month $m$. The choice rests on two properties of the shock, neither of which involves the macro variables. Aggregating a linear daily VAR to monthly frequency makes the monthly reduced-form innovation, to a first order, an average of the daily innovations, so the monthly structural shock is proportional to the average of the daily ones, the scheme also used by Alessandri et al. (2023); the *sum* differs from it only by the factor $D_m$, which the normalisation below removes up to its variation across months. The exact weights are triangular, since a shock early in the month affects more of the monthly average than one late in the month (Kilian, 2024), so (13) is the first-order filter; Online Appendix C.2

discusses the approximation. And an instrument must inherit zero mean and approximate serial uncorrelatedness, which only the signed mean does: the absolute value and the square destroy the sign, are not zero-mean, are strongly serially correlated because volatility clusters, and are not exogenous, since the intensity of one shock loads on the common volatility factor and hence on the other shocks. They are intensity measures, suitable for the level-on-level validation of Section 4.5 and never instruments, and Section 5.1 verifies this. One property of (13) must be declared: since $\mathrm{Var}(Z_m) = \sigma^2/D_m$ and $D_m$ ranges from 13 to 23 in this sample, the instrument is conditionally heteroskedastic by construction, independently of anything inherited from the daily shocks. The signs are also re-anchored once, at the daily level and before aggregation, so that all three shocks are adverse and the three response panels can be compared at a glance; Online Appendix C.3 records the convention and the one place where it changes a reported sign.

**The internal instrument.** Let $Z_t$ denote the aggregated shock for one of the $M$ daily shocks. The macro stage estimates the augmented system

$$\widetilde{X}_t = \left(Z_t,\ X_t'\right)', \qquad \widetilde{X}_t = \sum_{j=1}^{p_m} \widetilde{A}_j \widetilde{X}_{t-j} + \widetilde{A}_0 + \tilde{\varepsilon}_t, \tag{14}$$

with the instrument ordered first, and identifies recursively through $\widetilde{\Sigma} = PP'$ with $P$ lower triangular and $b = P_{\cdot,1}/P_{1,1}$. Because $Z_t$ is ordered first and is contemporaneously exogenous, the first Cholesky shock of the augmented system is the shock the instrument proxies, and the normalisation by $P_{1,1}$ expresses the response per unit movement of $Z$. This is the internal-instrument identification of Plagborg-Møller and Wolf (2021). Each shock is instrumented in its own augmented system, so no recursive ordering is imposed *among* the instruments; their mutual orthogonality is inherited from the daily stage, and Section 5.1 verifies that it survives aggregation. The system is estimated in levels under the hierarchical Minnesota prior (Litterman, 1986) of Giannone et al. (2015) at $p_m = 15$, with prior means set by integration order and the shrinkage hyperparameters chosen by maximising the marginal likelihood; Online Appendix C.4 gives the prior and the lag-order argument, and Section 5.2 reports the robustness check.

The internal estimator is preferred to the external proxy-SVAR for a reason that is binding in our framework rather than in principle: Plagborg-Møller and Wolf (2021) prove that under invertibility the two recover the same normalised responses, and that without invertibility the internal estimator still identifies them while the external one does not. Measurement error does not by itself separate the two; in the internal design it attenuates every response by a common factor, which the normalisation below removes. Section 5.1 shows that one of our three instruments is partly predictable from lagged real activity, and Table 11 documents serial correlation in another. What the augmented system identifies is visible in its first equation,

$$Z_t \ = \ c + \sum_{j=1}^{15} a_j' \widetilde{X}_{t-j} \ + \ \eta_t, \tag{15}$$

so the object entering the identification is $\eta_t$, the component of the monthly aggregate that fifteen months of the entire system's history could not have predicted. The identified shock is the unanticipated component of the monthly aggregate and not the aggregate itself. Any

predictable component is filtered out by the estimation rather than carried into the identification, and Section 5.1 draws the consequence for the economic label.

**Normalisation and inference.** Equation (14) returns responses per unit movement of $Z$, and $Z$ has no natural unit: its standard deviation is about 0.21, so one unit is a shock of roughly five standard deviations. Every reported response is rescaled by

$$c = \frac{\gamma}{\left|b_{\text{ref}}\right|}, \qquad \gamma = 0.25, \quad \text{ref} = \text{EBP}, \tag{16}$$

so that each shock is expressed as the shock that moves the excess bond premium by 25 basis points on impact. This is more than a choice of units. When the instrument is contaminated by measurement error unrelated to the shock of interest, which is the case that motivated the internal design, Plagborg-Møller and Wolf (2021) show that the estimator attenuates every response by a common factor $\lambda$, the signal-to-noise ratio of the instrument, identical at all horizons and for all variables. Scalings expressed in the instrument's own units leave $\lambda$ in the reported number and would systematically shrink whichever shock has the noisier instrument, whereas a ratio to a variable inside the system cancels it: the reported object is $\gamma\,\Theta_i/|\Theta_{\text{ref}}|$, which is the relative impulse response that the internal instrument identifies, and the unit effect normalisation of Stock and Watson (2018). The reference variable is chosen by a stated criterion rather than by preference, and Online Appendix C.6 reports it together with the alternatives.

Inference at the monthly stage is a residual-based moving-block bootstrap of the augmented VAR, valid under conditional heteroskedasticity and heavy tails (Brüggemann et al., 2016; Jentsch and Lunsford, 2019), with $\ell = 23$ months and $B = 2{,}000$ replications. Two implementation points matter enough to state here. The estimator replicated is the estimator reported, since writing the conjugate prior as dummy observations makes the posterior mean ordinary least squares on the augmented data, so every replication is shrunk exactly as the point estimate is. And the normalisation is applied inside each replication rather than to the finished bands, because the reported response is a ratio whose sampling distribution reflects uncertainty about the denominator; computing a band under one scaling and multiplying it by a factor read off the point estimate produces an interval for a different object (Stock and Watson, 2018). Online Appendix C.7 reports the dispersion of the scale factor and what the correction does to the bands. One limitation belongs here: the bands price uncertainty about the monthly VAR and its residual process, and do not propagate uncertainty about the daily identification, which is treated as the point-estimated shock series. The reported intervals are conditional on the daily rotation and understate total uncertainty by an amount this design does not measure.

# 4 The identified shocks

The identification returns three mutually orthogonal statistical objects, unique up to sign and permutation, with no economic content attached to any of them. This section reports what the daily stage delivers and then assigns the labels. Assigning them is a separate inferential step, taken after the rotation is fixed and feeding nothing back into it, and it rests on four readings kept independent of one another: the impact signature within the financial system, the shock's

own excess kurtosis, the dates of its largest realisations, and correlation with series that enter no estimation step. A label is warranted only where the four agree.

## 4.1 Diagnostics and the spectrum

The marginal likelihood selects $p^* = 9$ over a notably flat surface: the runner-up $p = 4$ is 2.04 log-units below the maximum and only $p = 3$ is decisively rejected, so it would overstate the evidence to claim that nine lags are strongly preferred to four (Online Appendix F.1). We keep the maximiser for two reasons. The criterion's answer is the criterion's answer, and the Occam factor has already priced the additional parameters. More substantively, a longer conditional mean is the conservative choice for this identification: the identifying signal is the fourth moment of the residuals, any conditional-mean dynamics left unmodelled reappear there as predictable variation, and the cost of removing more of that contamination is 28 coefficients per equation against 13 with 9,058 observations. At $p = 9$ the posterior mean of $\Sigma_u$ has condition number 75.0, against 348.7 for a extended seven-variable system and $1.12 \times 10^5$ with the raw VIX.

Breusch–Godfrey (Breusch, 1978; Godfrey, 1978) rejects in two of the three equations, with the largest statistic on CSPREAD (44.77), which is the equation with by far the highest unconditional kurtosis. The test is derived under conditional homoskedasticity and over-rejects under strong GARCH effects, so in daily financial data a rejection is a joint signal of possibly minor conditional-mean autocorrelation and typically dominant conditional heteroskedasticity. Neither is a defect a higher linear lag order repairs, and a lag-grid search up to $p = 25$ neither improves the marginal likelihood nor removes the autocorrelation in that equation. The resulting uncertainty is quantified by the block bootstrap instead.

**Table 2: Non-Gaussianity of the pre-whitened residuals.** $\boldsymbol{\iota}_t = \widehat{\Sigma}^{-1/2}\hat{u}_t$, baseline system, $p = 9$. Hill($\hat{\alpha}$) is the tail index computed on the upper five per cent of absolute residuals; KW.Exc is the octile-based robust excess kurtosis of Moors (1988) ($KR_2$ in Kim and White, 2004). Pre-whitening uses the posterior mean of $\Sigma_u$, so $\max|\mathrm{Cov}(\boldsymbol{\iota}) - I_M| = 3.29 \times 10^{-3}$ is the degrees-of-freedom factor $k/T_{\text{eff}}$; a common rescaling of $\boldsymbol{\iota}_t$ leaves the rotation unchanged.

| Variable | Kurt.Exc | Hill($\hat{\alpha}$) | Hill(right) | Hill(left) | KW.Exc |
|---|---|---|---|---|---|
| SP500 | 9.988 | 2.651 | 2.631 | 2.671 | 0.329 |
| VIX | 9.256 | 2.912 | 2.487 | 3.337 | 0.091 |
| CSPREAD | 34.323 | 2.377 | 2.323 | 2.432 | 0.249 |

Gaussianity is rejected decisively and on every measure (Table 2). CSPREAD has an excess kurtosis of 34.3, more than three times that of the other two components, which reflects the infrequent but very large discrete moves characteristic of credit markets. The Hill estimator (Hill, 1975) returns $\hat{\alpha} < 4$ for all three components in both tails; read literally this would imply a non-existent fourth moment, and Online Appendix E shows the literal reading is not supported, since a power law consistent with these indices would generate sample kurtoses between 1.8 and 2.9 times larger than those observed, whereas a finite-moment jump mixture matches both statistics jointly. The Hill values are retained as a measure of tail thickness and as a reason to bootstrap. The Kim–White octile statistic (Kim and White, 2004; Moors, 1988) is positive and significant for all three, with moving-block bootstrap 95% intervals of $[0.28, 0.38]$, $[0.05, 0.13]$

and $[0.18, 0.31]$: on this scale the bulk of the SP500 and CSPREAD distributions is as heavy as a Laplace, and that of the VIX as a Student-$t$ with five degrees of freedom. The leptokurtosis is therefore a property of the whole distribution, although most of the fourth-moment content, and hence of the identifying information, comes from the tails (Section 4.2).

**Independence.** Linear serial correlation in the identified shocks is mild, rejecting only for $v_1$ and only marginally. Volatility clustering rejects overwhelmingly for all three, with ARCH–LM statistics in the hundreds to thousands: the structural shocks are emphatically not i.i.d. As set out in Section 3.4 this does not bias the rotation, and it is why inference on the eigenvalues rests on the block bootstrap. The co-kurtosis rejects for all three pairs, with $\max_{i \neq j} |\mathbb{E}[v_i^2 v_j^2] - 1| = 4.30$; by (12) a rejection there was predicted rather than discovered, and its magnitude is an estimate of $\mathrm{Var}(\omega)$.

**Table 3: SV-robust contemporaneous independence diagnostics.** Moving-block bootstrap 95% confidence intervals, $\ell = 20$, $B = 5{,}000$. Both statistics have null value zero and are robust to a common stochastic-volatility factor by (12). The non-robust co-kurtosis panel and the full set of point estimates are in Online Appendix F.2.

| $i$ | $j$ | Estimate | Boot. SE | 95% CI | |
|---|---|---|---|---|---|
| *Panel A: co-skewness* $\mathbb{E}[v_i^2 v_j]$ | | | | | |
| $v_1$ | $v_2$ | −0.175 | 0.141 | [−0.464, 0.081] | accept |
| $v_1$ | $v_3$ | 0.104 | 0.102 | [−0.088, 0.316] | accept |
| $v_2$ | $v_1$ | −0.029 | 0.097 | [−0.206, 0.177] | accept |
| $v_2$ | $v_3$ | 0.315 | 0.090 | [ 0.162, 0.515] | **reject** |
| $v_3$ | $v_1$ | 0.080 | 0.050 | [−0.018, 0.179] | accept |
| $v_3$ | $v_2$ | 0.134 | 0.118 | [−0.078, 0.390] | accept |
| *Panel B: asymmetric fourth moment* $\mathbb{E}[v_i^3 v_j]$ | | | | | |
| $v_1$ | $v_2$ | 1.344 | 3.269 | [−4.857, 7.973] | accept |
| $v_1$ | $v_3$ | −0.544 | 1.864 | [−4.420, 3.124] | accept |
| $v_2$ | $v_1$ | −0.987 | 1.564 | [−3.953, 2.281] | accept |
| $v_2$ | $v_3$ | −1.391 | 0.803 | [−3.161, −0.028] | **reject** |
| $v_3$ | $v_1$ | −0.053 | 0.396 | [−0.927, 0.598] | accept |
| $v_3$ | $v_2$ | 2.215 | 1.265 | [ 0.228, 5.106] | **reject** |

The informative panels tell a different story from the co-kurtosis. Across the twelve SV-robust tests, nine accept the null, and all three violations involve the same single pair, $v_2$–$v_3$. Of the three distinct pairs in the system, two are clean on every SV-robust test. The violating pair is the equity and volatility components, and what it detects is the leverage effect, the documented asymmetry whereby equity declines raise implied volatility more than equity gains lower it. That is a known feature of the data, it is economically interpretable, and a linear contemporaneously-independent factor structure cannot accommodate it. The honest summary is that Assumption 1(ii) holds approximately and not exactly, that the single documented violation is the leverage effect, and that the credit shock is independent of both other components on every SV-robust test.

The spectrum decays sharply and monotonically (Table 4). The leading shock carries an excess kurtosis more than three times that of the second and more than four times that of

**Table 4: The spectrum and its separation.** Panel A: eigenvalues $d_i$ of $K(Z)$ and realised excess kurtosis $\kappa_i = \mathbb{E}[v_i^4] - 3$ of the identified shocks, posterior median and $[16, 84]$ percentiles over 5,000 draws, with moving-block bootstrap 95% interval for $d_i$ ($B = 500$, $\ell = 20$). Panel B: near-degeneracy test on adjacent pairs; $\Pr^*$ is the bootstrap frequency of the same ordering, with each resampled eigenvalue matched to its reference direction. Under Assumption 1 the two columns of Panel A estimate the same object by different routes.

| *Panel A: spectrum* | | | | |
|---|---|---|---|---|
| Shock | $d_i$ (median) | $d_i$ boot. 95% CI | $\kappa_i$ (median) | $\kappa_i$ $[16, 84]$ |
| $v_1$ | 42.61 | [ 6.60, 98.64] | 37.80 | [36.17, 39.46] |
| $v_2$ | 18.64 | [ 5.11, 34.76] | 12.38 | [11.82, 12.96] |
| $v_3$ | 11.09 | [ 4.74, 20.44] | 8.63 | [ 8.22, 9.06] |
| *Panel B: near-degeneracy* | | | | |
| Pair | $\Pr(\lambda_i > \lambda_{i+1} \mid \mathbf{Y})$ | $\mathrm{med}(\Delta\lambda)$ | $q_{16}(\Delta\lambda)$ | $\Pr^*(\lambda_i > \lambda_{i+1})$ |
| $v_1$–$v_2$ | 1.0000 | 23.98 | 22.27 | 0.968 |
| $v_2$–$v_3$ | 1.0000 | 7.55 | 6.83 | 0.802 |

the third. Posterior bands are tight and bootstrap bands are far wider: for $v_1$ the bootstrap interval spans more than twenty times the posterior band, which is the inflation of the sampling variance of $\widehat{K}$ that volatility clustering predicts and the reason both are reported. Panel B is the empirical verification of Assumption 1(iii): both adjacent pairs are strictly ordered with posterior probability one and, more demandingly, the 16th percentile of each gap is comfortably positive. No pair is near-degenerate under the posterior, so no subspace exists within which the identified directions are only jointly determined. The concentration ratios sharpen this. The eigenvalue ratio $\lambda_1/\lambda_2$ has posterior median 2.288 with $\Pr(\lambda_1/\lambda_2 > 2 \mid \mathbf{Y}) = 0.998$, while the economically relevant ratio of the true kurtoses has median 3.054 with $\Pr(\kappa_{(1)}/\kappa_{(2)} > 2 \mid \mathbf{Y}) = 1.000$. That this statement can be made at all follows from the identity weight, whose eigenvalues are the kurtoses themselves.

The block bootstrap is the more demanding test, because it resamples the tail observations themselves, and it is the one to rely on where the two disagree. Each resampled eigenvalue is matched to its reference direction before any comparison is made, so the frequencies below refer to identified directions and are not true by construction. The credit direction keeps the largest eigenvalue in 96.8 per cent of the resamples, with $\Pr^*(\lambda_1/\lambda_2 > 2) = 0.778$, whereas the equity and volatility eigenvalues keep their order in 80.2 per cent, below the 0.95 threshold used for the posterior flag. The leading shock is therefore well separated from the other two, while the ordering of $v_2$ and $v_3$ is less secure. The directions themselves are stable. The column-instability indices are 0.038, 0.081 and 0.058, against about 0.36 if a pair of directions were degenerate and free to rotate within its plane, so all three directions survive resampling of the tail observations and the credit direction is the most stable of the three. The weaker separation between $v_2$ and $v_3$ affects their ordering, which no downstream object uses, more than their directions.

The two columns agree exactly in ordering but not in level, and the gap has an exact decomposition: in the structural basis

$$d_i \;=\; \kappa_i + \sum_{j \neq i} \Big( \mathbb{E}[v_i^2 v_j^2] - 1 \Big), \tag{17}$$

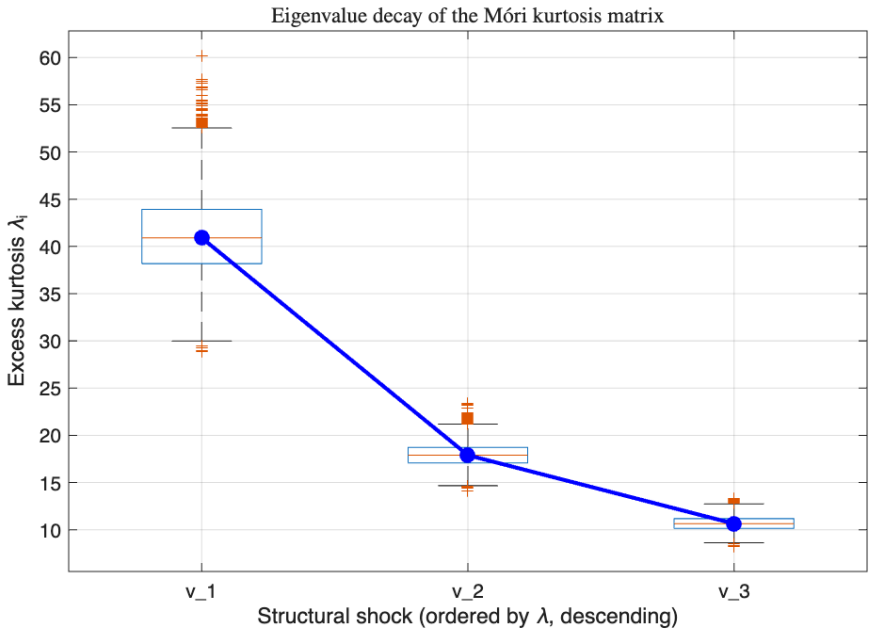



**Figure 1: Eigenvalue decay of the Móri kurtosis matrix.** Box plots show the posterior distribution of each $\lambda_i$ over 5,000 draws; the solid line traces the posterior median.

which the estimates reproduce once the small degrees-of-freedom rescaling of the whitened residuals noted under Table 2 is taken into account (Online Appendix B.6). The gap is therefore the excess co-kurtosis of Online Appendix F.2, which by (12) measures the common volatility factor and not a failure of Assumption 1(ii). Because that factor leaves the Móri matrix diagonal in the structural basis (Section 3.4), it shifts the eigenvalues without rotating the eigenvectors; ordering by the realised $\kappa_i$ or by $d_i$ gives the same ranking here.

## 4.2 The impact matrix and the labels

The reference rotation is computed at the posterior mean of $\Sigma_u$ with the sign convention of (11). The anchoring variable of each shock, with its standardised impact loading, is $v_1 \to$ CSPREAD (0.950), $v_2 \to$ SP500 (0.728), $v_3 \to$ VIX (0.992). The anchors are unambiguous: each shock has one variable that clearly dominates its standardised impact, and the three anchors are three different variables.

**Table 5: Structural impact matrix** $B^{(0)} = \widehat{\Sigma}^{1/2} P_{\text{ref}}$, posterior median. Entry $(m, k)$ is the contemporaneous response of variable $m$ to a unit-variance shock $v_k$, in original units. Eight of the nine signs have posterior agreement of at least 84%; the exception is the credit-spread response to $v_3$, which is essentially zero in magnitude. The reference rotation $P_{\text{ref}}$ is in Online Appendix F.2.

| | $v_1$ | $v_2$ | $v_3$ |
|---|---|---|---|
| SP500 | −0.0031 | 0.0083 | −0.0071 |
| VIX | 0.0050 | −0.0071 | 0.0666 |
| CSPREAD | 0.0194 | 0.0064 | −0.0001 |

Table 5 admits an immediate economic reading, obtained without a single restriction. Shock $v_1$ widens the quality spread by 1.9 basis points on impact, lowers equity and raises implied

volatility, a sign pattern $(-,+,+)$ on (SP500, VIX, CSPREAD) that is the canonical financial-stress signature. Shock $v_2$ raises equity by 0.83% and lowers volatility, the risk-on configuration, while mildly widening the spread. Shock $v_3$ raises implied volatility by 6.7 log points, by an order of magnitude the largest single entry in $B^{(0)}$, lowers equity and leaves the credit spread untouched at $-0.0001$. Scoring each shock against the canonical stress signature over the three variables available to test it gives 3 for $v_1$, 1 for $v_2$ and 2 for $v_3$: the credit shock attains the maximum and the equity shock scores lowest, since its risk-on configuration is the mirror image of the stress pattern. That checklist is the weakest of the readings used here. It has little resolution, cannot weight variables by the tail mass they carry, reads only signs while discarding magnitudes, and one of its three entries is fixed by construction because CSPREAD is also the sign anchor of $v_1$.

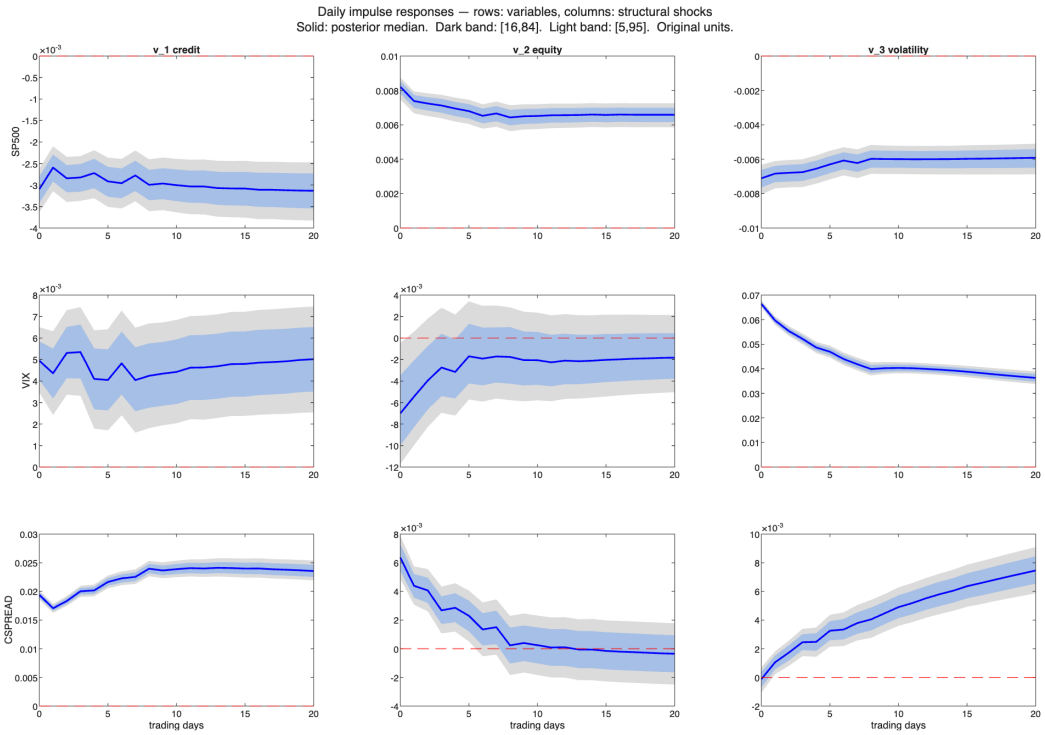



**Figure 2: Daily impulse responses to the three structural shocks.** Rows are variables, columns are shocks, original units. Solid line: posterior median; dark band $[16, 84]$; light band $[5, 95]$.

**Four readings, one object.** The labels rest on four criteria that can in principle disagree and that fail in different ways when identification is fragile. They select the same leading object.

1. **Own non-Gaussianity.** $v_1$ has the largest excess kurtosis, 37.80, more than three times 12.38 (Table 4).

2. **Separation from the runner-up.** $\Pr(\kappa_{(1)}/\kappa_{(2)} > 2 \mid \mathbf{Y}) = 1.00$, and the moving-block bootstrap keeps $v_1$ first in 96.8 per cent of resamples under serial dependence.

3. **Canonical sign signature.** $v_1$ scores 3/3 on the stress checklist, the maximum attainable in this system.

4. **Narrative dates.** The largest realisations of $v_1$ coincide with the recognised credit dislocations (Section 4.4).

The four are logically independent: the first reads a single eigenvalue, the second a posterior probability on their ratio, the third only the signs of $B^{(0)}$, and the fourth nothing about the model at all, only the calendar. Their agreement is what licenses the labels.

A fifth check is stronger than any of them, because it does not use the full sample. Re-running the identification on 2007–2009 and on 2019–2021, two non-overlapping three-year windows twelve years apart with no observation in common and neither informed by the other, returns leading directions of $(-0.494, -0.110, 0.863)$ and $(-0.227, -0.171, 0.959)$. The angle between them is 16.7 degrees, both anchor on CSPREAD, and the leading excess kurtosis is 24.2 and 45.9 respectively. The credit direction is recovered twice from disjoint samples using nothing but the fourth moments of three daily prices. The premise that makes this informative is that the tail mass is extraordinarily concentrated: the 85 trading days of October–November 2008 and March–April 2020, 0.9 per cent of the sample, carry 61 per cent of the fourth-moment content of the pre-whitened residuals. An estimator built on excess kurtosis reads those episodes by construction, so the question worth asking is not whether the rotation survives deleting the information that identifies it, but whether it rests on more than one episode. It does. Online Appendix F.3 reports the subsample table and the levels-versus-differences check.

**The contraction decides the ranking.** Lemma 1 is not an abstract statement in this sample. Table 6 evaluates the factor $c_r$ under the two weights used in this literature.

**Table 6: The two contractions evaluated in the data**, at the reference identification. $d_i$ is the eigenvalue of the Móri matrix $K(Z) = I \star c_4$, $\kappa_i$ the realised excess kurtosis of the identified shock, $s_i = \mathbf{1}'w_i$ the column sum of the rotation, and $\kappa_i s_i^2$ the eigenvalue the Kollo matrix $C(Z) = \mathbf{1}\mathbf{1}' \star c_4$ would assign under Assumption 1. Ranks in brackets. Under exact independence both weights return the same eigenvectors, hence the same three shocks; under a common volatility factor only the Móri weight does (Online Appendix D.2).

| Shock | $d_i = K(Z)$ eig. | $\kappa_i$ (realised) | $s_i$ | $\kappa_i s_i^2 = C(Z)$ eig. |
|---|---|---|---|---|
| $v_1$ (credit) | 42.446 [1] | 37.688 [1] | 0.492 | 9.105 [2] |
| $v_2$ (equity) | 18.493 [2] | 12.311 [2] | 1.657 | **33.799** [1] |
| $v_3$ (volatility) | 11.031 [3] | 8.632 [3] | 0.114 | 0.113 [3] |

The credit shock has the largest excess kurtosis in the system by a factor of three, and the smallest projection on the all-ones direction, $s_1 = 0.49$. The Kollo weight therefore assigns it 9.10 and demotes it to second, while promoting the equity shock, whose kurtosis is less than a third as large but whose projection is $s_2 = 1.66$, to first with 33.80. Ordering by that eigenvalue would lead one to call the equity shock the leading, most non-Gaussian shock when it is nothing of the sort, and the reversal is not marginal: the ratio $\kappa_1/\kappa_2 = 3.06$ becomes $33.80/9.10 = 3.71$ in the opposite direction. The factor $s_r$ has a concrete reading, how strongly a shock moves all variables in the same direction, which invites the thought that $C(Z)$ ranks the systemic factor first. The economics does not support it. A financial crisis moves variables with mixed signs, equity down and volatility and spreads up, so the naive sum cancels the equity decline against the other two instead of accumulating them, and in these data the shock with the largest $s_r$

is the one whose signature is risk-*on*. The received view that the all-ones weight is preferable for its generality is mistaken: the two estimators impose identical assumptions and recover the same shocks in population, and the extra factor is a contaminant that happens to break ties in a degenerate case that does not arise here. Online Appendix D develops the argument and places both in the ICA literature, where a single contraction is a computational shortcut and joint diagonalisation (Cardoso and Souloumiac, 1993) is the principled answer to robustness.

## 4.3 Variance and kurtosis answer different questions

**Table 7: Daily forecast error variance decomposition**, posterior medians. Rows sum to one up to Monte Carlo error, since medians are not additive. Dominant attribution per row in **bold**.

| | $v_1$ (credit) | $v_2$ (equity) | $v_3$ (volatility) |
|---|---|---|---|
| *Panel A: $h = 1$* | | | |
| SP500 | 0.0697 | **0.5200** | 0.4101 |
| VIX | 0.0054 | 0.0100 | **0.9843** |
| CSPREAD | **0.9154** | 0.0823 | 0.0017 |
| *Panel B: $h = 20$* | | | |
| SP500 | 0.0956 | **0.4938** | 0.4096 |
| VIX | 0.0109 | 0.0046 | **0.9834** |
| CSPREAD | **0.9431** | 0.0095 | 0.0466 |

The decomposition is close to block-diagonal and remarkably stable across horizons. Each market segment has a clearly dominant owner: the credit spread belongs to $v_1$ (92–95%), the VIX to $v_3$ (98%), and equity to $v_2$ (49–52%) with $v_3$ a strong secondary contributor at 41%, which is the leverage effect expressing itself through the structural decomposition.

Table 7 should not be read as a ranking of importance. On the second moment the credit shock is a specialist: it dominates one variable and explains 7.0% of equity variance and 0.5% of volatility variance, which is a shock one might reasonably call narrow. On the fourth moment the same shock is the most non-Gaussian component of the system by a factor of three. Neither number is wrong, and they answer different questions. The variance decomposition asks how much of the typical fluctuation a shock accounts for, and by that standard the credit shock is modest outside its own market. Excess kurtosis asks how heavy the shock's own tails are, and by that standard it is dominant. Credit dislocations are rare, so in normal times the credit channel contributes little to the system's variation; when it moves, it moves violently, and a fourth-moment measure registers exactly that. For a question about financial stress, which is an inherently tail phenomenon, the second measure is the relevant one, and this dissociation is what a covariance-based identification cannot see.

## 4.4 Narrative validation

An identification resting entirely on higher moments, with no economic restriction imposed anywhere, invites an obvious test: do the identified shocks fire when economic history says they should? No date information enters any estimation step.

**Table 8: The ten largest absolute realisations of each structural shock**, daily frequency.

| $v_1$ (credit) | | $v_2$ (equity) | | $v_3$ (volatility) | |
|---|---|---|---|---|---|
| Date | $\lvert v_1 \rvert$ | Date | $\lvert v_2 \rvert$ | Date | $\lvert v_3 \rvert$ |
| 2020-03-23 | 19.50 | 2020-03-24 | 10.52 | 2018-02-05 | 12.52 |
| 2008-10-15 | 17.86 | 2020-03-20 | 10.09 | 2024-12-18 | 8.90 |
| 2020-04-15 | 12.08 | 2008-10-13 | 9.94 | 2024-08-05 | 8.34 |
| 2008-11-20 | 10.86 | 2008-10-28 | 9.60 | 2007-02-27 | 7.63 |
| 2008-10-17 | 9.57 | 2020-03-16 | 9.47 | 2021-01-27 | 7.57 |
| 2020-03-17 | 9.18 | 2008-10-09 | 8.20 | 2021-11-26 | 6.79 |
| 2008-10-10 | 7.57 | 2009-03-23 | 7.88 | 2025-04-04 | 6.56 |
| 2008-11-18 | 7.23 | 2020-03-12 | 7.77 | 2015-08-24 | 6.51 |
| 2020-03-20 | 7.21 | 2020-03-26 | 7.70 | 1990-07-23 | 6.43 |
| 2001-12-11 | 7.15 | 2020-04-06 | 7.70 | 2011-08-08 | 6.32 |

The validation is strong, and strongest where it matters most. Nine of $v_1$'s ten largest realisations fall in two windows, March–April 2020 and October–November 2008. The largest is 23 March 2020, the day the Federal Reserve announced unlimited quantitative easing and the corporate credit facilities in response to a credit market that had ceased to function; the second is 15 October 2008, at the peak of the post-Lehman funding freeze; the tenth, 11 December 2001, falls in the Enron aftermath. No date in the column is a pure equity or pure volatility episode.

The volatility column is the sharper result, because it is the one that could most easily have failed. Its dates are the collapse of the inverse-VIX exchange-traded products (5 February 2018), the hawkish FOMC projections that repriced the 2025 easing path (18 December 2024), the unwinding of the yen carry trade (5 August 2024), the Shanghai correction (27 February 2007), the GameStop episode (27 January 2021), the Omicron announcement (26 November 2021), the tariff sell-off (4 April 2025), the Chinese devaluation flash crash (24 August 2015), 23 July 1990, and the S&P downgrade of U.S. sovereign debt (8 August 2011). Nine of the ten are violent spikes in implied volatility unaccompanied by any credit dislocation, and the exception is 23 July 1990 in the opening months of the sample, for which we offer no clean reading. What the labels turn on is the converse, and it holds without exception: not one of these dates is a credit crisis, and not one of the canonical credit dates appears among them. The identification separated volatility events from credit events using nothing but fourth moments. The equity column concentrates in the same two crises but on different days, and adds 23 March 2009, the announcement of the Public–Private Investment Program and one of the largest single-day rallies of the period, which is a strongly positive equity event and what one expects of a risk-appetite shock whose positive direction is risk-on. Across the three columns only 20 March 2020 recurs, shared by $v_1$ and $v_2$, and the volatility column shares nothing with either of the others.

The same check can be run on the *monthly* object. From here to the end of the paper the shocks appear in aggregated form: the daily series $v_{k,t}$ are collapsed to monthly frequency by the signed mean (13) of Section 3.5, which is the filter a linear VAR requires and the object Section 5 uses as its instrument. Averaging some twenty-one trading days could in principle blur three components into a single measure of financial turbulence, so the separation has to be verified and not assumed. It survives intact: no month appears in two columns, the credit column is the post-Lehman freeze and the COVID credit seizure, the volatility column is

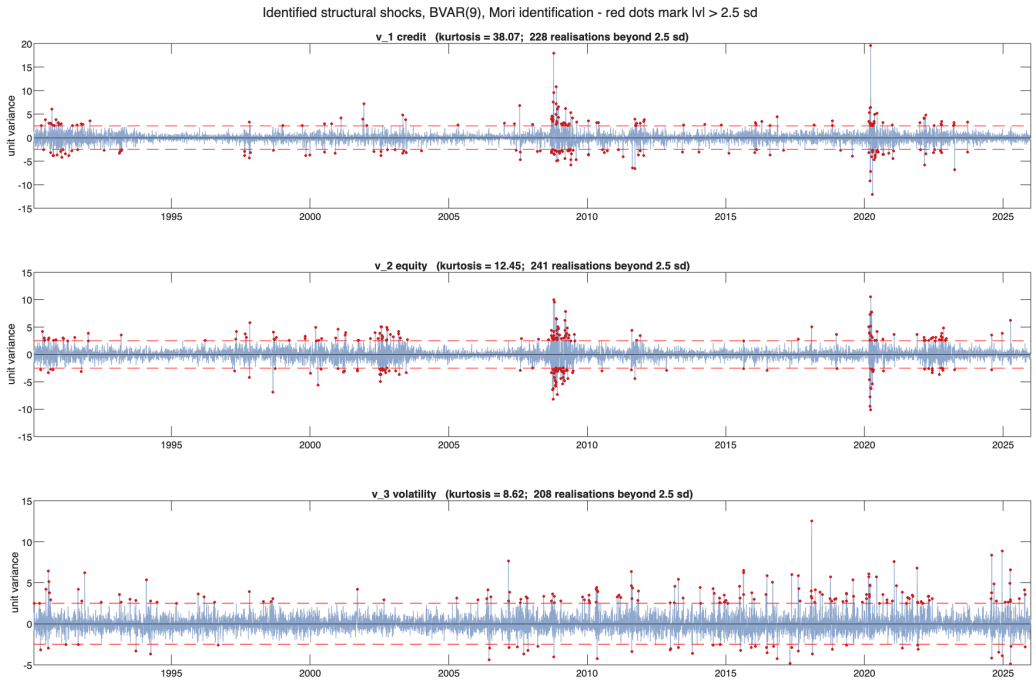



**Figure 3: The three identified structural shocks**, unit variance, one panel per shock, with the $\pm 2.5$ standard-deviation thresholds dashed and the realisations beyond them marked. Each panel is headed by the shock's realised excess kurtosis and by the number of such realisations.

LTCM, September 2001, the Flash Crash, the Chinese devaluation and the onset of COVID, and the equity column is the dot-com peak, the 2002 trough and the turn of early 2009 (Online Appendix F.4). Two features of that record go beyond the labels. And aggregation treats the three shocks differently. Under independence, averaging $n$ observations divides excess kurtosis by $n$, so at $n \approx 21$ one would predict a retention of some five per cent; the equity and volatility shocks retain ten and twelve per cent, consistent with the GARCH-driven slowdown of Drost and Nijman (1993), whereas the credit shock retains fully half and its monthly aggregate remains strongly right-skewed. Volatility spikes are single days, credit dislocations are month-long episodes, so averaging within the month reinforces the credit shock instead of diluting it. The component that is most non-Gaussian at daily frequency is also the one whose non-Gaussianity best survives the bridge.

**Historical decomposition.** Table 9 turns the exercise around and asks which shock moved which variable on a given date, reporting the five-day cumulative effect of the event-day shock at Lehman Brothers, the COVID-19 crash and the Silicon Valley Bank failure. A decomposition entry is an impulse response multiplied by the *realised* value of the shock on the date, so a shock that realises negatively has its whole column mirrored.

Crises share a skeleton and differ in one limb. The credit spread is a credit-shock phenomenon in every episode, with $v_1$ accounting for $+0.37$, $+0.29$ and $+0.13$ percentage points over five days against contributions from the other two an order of magnitude smaller and often of opposite sign, and implied volatility is a volatility-shock phenomenon in the same sense, with $v_3$ delivering

**Table 9: Historical decomposition at three crisis dates.** Five-day cumulative effect of the event-day shock, original units. Day-zero contributions to the forecast error are in Online Appendix F.5.

| Event | Variable | $v_1$ | $v_2$ | $v_3$ |
|---|---|---|---|---|
| Lehman | SP500 | −0.0550 | −0.0700 | −0.0943 |
| | VIX | 0.0934 | 0.0425 | 0.7844 |
| | CSPREAD | 0.3702 | −0.0386 | 0.0212 |
| COVID | SP500 | −0.0426 | −0.3511 | −0.1586 |
| | VIX | 0.0722 | 0.2128 | 1.3198 |
| | CSPREAD | 0.2863 | −0.1934 | 0.0357 |
| SVB | SP500 | −0.0196 | −0.0027 | −0.0505 |
| | VIX | 0.0332 | 0.0016 | 0.4201 |
| | CSPREAD | 0.1316 | −0.0015 | 0.0114 |

+0.78, +1.32 and +0.42. The equity decline is where the episodes genuinely differ. At COVID it is overwhelmingly the equity shock (−0.351 of −0.552), consistent with a broad repricing of the market; at SVB the equity shock is essentially absent (−0.003 of −0.073) and the decline is a by-product of the volatility spike, consistent with a banking and duration event that widened spreads and raised volatility without repricing equity generally; at Lehman it is spread across all three. What distinguishes one crisis from another is how much of the equity decline is a repricing of risk appetite and how much is a spillover from volatility.

## 4.5 The external validation and the bridge

This subsection validates them against independently constructed series that no part of the estimation has seen: the news-based Economic Policy Uncertainty indices of Baker et al. (2016) for the United States and the two global variants of Davis (2016), and the Geopolitical Risk indices of Caldara and Iacoviello (2022) in a global and a US-specific variant (Online Appendix A.8). A correlation appearing here cannot have been engineered by the specification.

The exercise is conducted entirely at monthly frequency and after aggregation, and because that order decides how the results read it is worth stating. Two monthly objects are built from the daily shocks. The first is the signed mean (13) of Section 3.5, the same aggregate that Section 5 uses as its instrument, which preserves the sign and asks in which direction the shock moved. The second is an intensity aggregate $\text{abs}_k = \sum_{t \in m} |v_{k,t}|$, built only for this subsection, which discards the sign and records how turbulent the month was. The five indices are then read at their published monthly frequency, matched to the shock aggregates on the calendar month, and correlated on those matched pairs. Nothing here is computed at daily frequency, and the intensity aggregate is never used as an instrument, being neither zero-mean nor serially uncorrelated (Section 5.1).

A structural shock is unforecastable at $t-1$ and therefore has no reason to co-move with the predictable part of a persistent index, only with its innovation. The customary remedy, the monthly log-change, imposes a unit root on indices that are persistent without being integrated, which over-differences and attenuates the correlation towards zero. We estimate the persistence instead, taking as the unforecastable component the residual of an AR(6) on the log level. Two

pairings follow and cannot be mixed. The signed design matches the monthly mean against that innovation and asks whether the shock moves in the direction uncertainty moves; both objects are innovations, and it carries the inferential weight. The amplitude design matches an intensity aggregate against the absolute innovation and asks whether the shock is more violent when uncertainty is high. Online Appendix F.6 reports the third, descriptive pairing on levels, the side-by-side comparison of the two filters, and the full set of forty-five coefficients.

**Table 10: Correlation of the monthly-aggregated shocks with external uncertainty indices.** $\hat{\eta}$ is the AR(6) innovation of the index. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Each coefficient is computed on the intersection of the index's own support with the shock sample, so the effective window differs across rows. Signs in Panel A follow the bridge convention under which all three shocks are adverse; Panel B is invariant to it.

| | $v_1$ (credit) | $v_2$ (equity) | $v_3$ (volatility) |
|---|---|---|---|
| *Panel A: signed, index innovation* $\hat{\eta}$ | | | |
| US_EPU | 0.088* | 0.066 | **0.238***** |
| GEPU_current | 0.086 | 0.131** | **0.204***** |
| GPR | 0.007 | **0.148***** | 0.068 |
| GPRC_USA | −0.006 | **0.136***** | 0.074 |
| *Panel B: amplitude,* $\lvert\hat{\eta}\rvert$ *against shock intensity* | | | |
| US_EPU | 0.217*** | 0.226*** | 0.199*** |
| GEPU_current | 0.210*** | 0.187*** | 0.200*** |
| GPR | 0.046 | 0.054 | 0.117** |
| GPRC_USA | 0.047 | 0.045 | 0.124** |

The volatility shock is the component most closely associated with measured uncertainty. In the signed design it moves with policy uncertainty and carries the largest coefficients in the panel, so the component that the impact matrix labels "volatility" on the strength of a 6.7 log-point loading on the VIX is the same one that rises when an independently constructed, text-based measure of policy uncertainty rises, with nothing in the estimation connecting the two. The equity shock passes a test that could genuinely have failed: under the bridge convention $v_2$ is the shock that lowers equity, and it correlates positively with every index, which is the sign an adverse equity shock is required to have and which the data were free to refuse. The credit shock relates to uncertainty in amplitude far more than in direction, with a signed association roughly a third of the volatility shock's and the strongest amplitude association of the three. Credit dislocations are more violent when policy uncertainty is high without having much systematic direction with respect to it, which is the same picture that an excess kurtosis of 37.8 and month-long episodes give.

The two panels together sharpen this into something the labels depend on. Panel B is nearly uniform across shocks against the EPU family, all three lying between 0.17 and 0.23 and statistically indistinguishable, whereas the same three shocks return 0.088, 0.066 and 0.238 in Panel A. Policy uncertainty is associated with the turbulence of every financial component in roughly equal measure, and what separates the components is the direction in which they move with it, a discrimination that lives entirely in the signed design. Geopolitical risk supplies the discriminating null. It is uncorrelated with the amplitude of the credit and equity shocks, which is reassuring rather than disappointing, since a design returning significant correlations

everywhere would be diagnosing spurious common trending; where it does register is on the equity shock in the signed design, so geopolitical events move equity risk appetite and say nothing about credit. Two caveats govern how much these correlations can bear. The table reports forty-five coefficients in total without multiplicity correction, so two or three spurious rejections would be expected at the nominal five per cent level, and what matters is the structure across columns and not any individual entry. And the descriptive level-on-level pairing reported in Online Appendix F.6 relates two persistent series under $p$-values that assume i.i.d. sampling, which is why the ordering it produces is confirmed here on filtered series before being used.

# 5 Macroeconomic transmission

The daily analysis identifies which shock generates the tails of the financial system defined in the BVAR. This section asks whether that matters for the real economy. The three daily shocks are aggregated by (13), re-anchored so that all three are adverse, and used as internal instruments in the ten-variable BVAR of (2), estimated on 429 months from January 1990 to September 2025 at $p_m = 15$. All responses are expressed per shock moving the excess bond premium by 25 basis points on impact. Bands are the moving-block bootstrap of Section 3.5, computed on the same shrunk estimator that produces the reported line and with the normalisation applied inside every replication, so each interval is an interval on the reported ratio.

## 5.1 Relevance, whiteness and predictability

Section 3.5 argues that only the signed monthly mean inherits the properties that make an object a shock. The argument is testable, and Table 11 tests it.

**Table 11: Serial correlation of the monthly aggregates**, 432 months. $\rho(1)$ is the first-order autocorrelation and LB(12) the Ljung–Box statistic on twelve monthly lags. The signed mean is the instrument; the intensity aggregate is shown for comparison and is used only in the validation of Section 4.5. The squared aggregates are in Online Appendix F.7.

| Shock | Aggregate | $\rho(1)$ | LB(12) | $p$-value |
|---|---|---|---|---|
| $v_1$ credit | signed mean | 0.169 | 34.5 | 0.0006 |
| | intensity | 0.674 | 615.4 | 0.0000 |
| $v_2$ equity | signed mean | −0.001 | 24.0 | 0.0202 |
| | intensity | 0.627 | 484.5 | 0.0000 |
| $v_3$ volatility | signed mean | −0.038 | 19.5 | 0.0769 |
| | intensity | 0.292 | 160.2 | 0.0000 |

The intensity measures fail decisively, as they must. Volatility clusters, so any non-negative function of the shock's amplitude inherits that clustering, with first-order autocorrelations up to 0.674 and statistics an order of magnitude beyond any conventional critical value. This turns the argument of Section 3.5 into a measurement: these objects are persistent state variables and not shocks, and using them as instruments would mix contemporaneous with lagged effects. The signed means behave very differently, with first-order autocorrelations of at most 0.169 in absolute value. Among them the volatility instrument is not rejected at five per cent, the equity

instrument is rejected at five but not at one on a first-order autocorrelation of −0.001, and the credit instrument is rejected decisively.

The source of that rejection appears to be the underlying price series and not the identification. The Moody's seasoned yields from which CSPREAD is built are computed from a narrow panel with matrix pricing, so many constituent bonds do not trade on a given day and their prices are interpolated; the resulting daily series is mechanically smooth, and a shock extracted from it inherits that smoothness. The diagnostic evidence supports this reading, since the Breusch–Godfrey statistic is largest for the CSPREAD equation and extending the daily lag order removes neither. What matters for the estimator is that the internal design identifies the innovation $\eta_t$ of (15) and not $Z_t$ itself, so a first-order autocorrelation of 0.169 in the level is filtered out by the estimation.

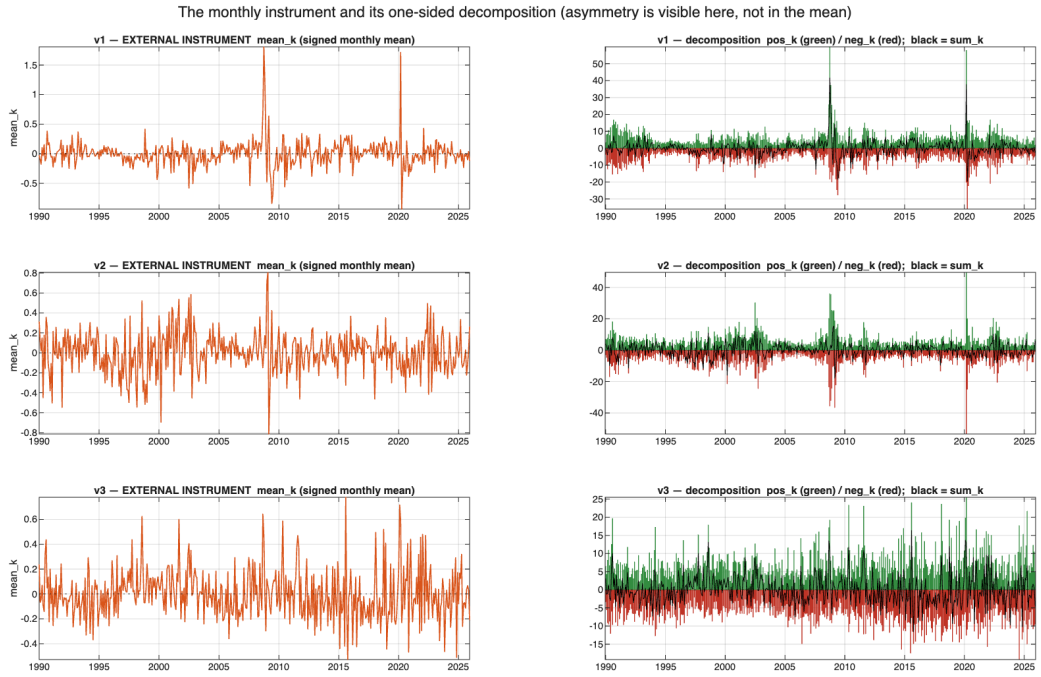



**Figure 4: The monthly instrument and its one-sided decomposition**, one row per shock. Left column: the signed monthly mean $Z_m^{(k)}$ of (13), the object used as the instrument, fluctuating around zero without a discernible low-frequency component. Right column: the same month split into its positive and negative parts, where the asymmetry of each shock becomes visible and the signed mean is silent.

Against the rule-of-thumb value of 10 all three instruments are relevant for the normalisation reference. Against the stricter critical value of Montiel Olea and Pflueger (2013) for a worst-case bias of 10 per cent at the 5 per cent level, 23.1 with a single instrument, only the volatility instrument passes. The design handles it by normalising inside every bootstrap replication, where the scale factor never changes sign in more than 0.2 per cent of draws, and the variance decomposition, which carries the main quantitative claim, is invariant to the normalisation and therefore unaffected (Online Appendix C.8). The instruments are built from three daily price series by an identification that uses no monthly information and imposes no economic restriction, and they are nonetheless strongly correlated with the monthly innovation to an

**Table 12: First-stage relevance of the aggregated daily shocks.** $F_{\text{eff}}$ is the HC1-robust effective $F$ of Montiel Olea and Pflueger (2013). The EBP column is the statistic on the normalisation reference; the rule-of-thumb value is 10 and the Montiel Olea and Pflueger (2013) critical value for a 10% worst-case bias at the 5% level is 23.1; the next column is a *selected* maximum over the ten macro variables and is shown only to identify where each shock lands hardest.

| Shock | Daily anchor | sd($Z$) | $F_{\text{eff}}$ (EBP) | $\max_n F_{\text{eff}}$ (variable) | $F_{\text{eff}}$ (INDPRO) |
|---|---|---|---|---|---|
| $v_1$ | CSPREAD | 0.228 | 11.5 | 12.5 (RAI_BEX) | 0.51 |
| $v_2$ | SP500 | 0.207 | 14.2 | 14.2 (EBP) | 0.07 |
| $v_3$ | VIX | 0.203 | 21.6 | 72.2 (RAI_BEX) | 0.00 |

independently constructed risk-premium index. The excess bond premium of Gilchrist and Zakrajšek (2012) is estimated from bond-level micro data by different authors from different sources, so a fourth-moment rotation of daily prices predicting its monthly innovation is external validation of the identification. The three instruments also remain close to mutually orthogonal after aggregation, with pairwise correlations of $-0.035$, $+0.100$ and $+0.029$. Orthogonality at daily frequency comes from the independence-based identification and is nowhere imposed at the monthly stage, so its survival across the bridge is informative: the three instruments are not competing for the same monthly variation.

The effective $F$ against industrial production is indistinguishable from zero, and it would be a misreading to record this as a weak-instrument problem. A financial disturbance does not move physical production within the calendar month in which it occurs, production being a flow largely realised by the time the disturbance is observed, so a valid financial instrument should be expected to have no contemporaneous explanatory power for it. The informative statistic is the one computed against the variable through which the shock enters the system, and there the instruments are strongly relevant. The near-zero figure records the architecture of a two-stage design: the shock lands on financial prices and reaches the real economy through the estimated dynamics of the VAR, exactly as in Gertler and Karadi (2015), where the monetary surprise identifies the impact on the one-year rate and output responds only through the system. Section 5.3 makes that chain visible as a number. Where the weak-instrument concern does bite is in the *normalisation*, whose denominator is the impact on the reference variable, estimated with an effective $F$ between 11.5 and 21.6; the design responds by applying the normalisation inside every bootstrap replication, so the reported band is a band on the ratio and prices the denominator's sampling variability. No replication set shows a sign flip in the denominator in more than 0.2 per cent of draws, so the Cauchy-tail pathology of weak-instrument ratios does not arise here (Online Appendix C.7).

**Predictability.** A structural shock must be unpredictable from information available at $t-1$. Testing against the daily VAR's own variables would be vacuous, so each instrument is regressed on two lags of industrial-production growth, the change in the unemployment rate, PCE inflation and the excess bond premium, with an HC1-robust Wald statistic. The equity and volatility instruments pass ($p = 0.55$ and $p = 0.052$); the credit instrument does not, although the rejection is marginal, significant at five per cent and not at one, and the regression accounts for ten per cent of the variation in the monthly aggregate. One caveat cuts towards rejection and should

be recorded: the regressors are final-vintage series, whereas the shocks are built from prices observed in real time. Online Appendix C.9 sets out the test, and Table F.14 reports the full output.

Reading the individual coefficients is what makes the exercise informative. The two lags of the excess bond premium carry opposite signs (+1.99 and −2.02) and largely offset, which is the signature of mean reversion in the regressor and not of predictive content: what those terms detect is that credit spreads revert. The substantive part is elsewhere. The second lags of real activity, $\Delta \ln \text{INDPRO}(-2)$ and $\Delta\text{UNRATE}(-2)$, are both significant and both negative, so the credit shock is more likely to be adverse after two months of weakening activity. That constrains the label. What (15) delivers is the component of the monthly credit aggregate that fifteen months of the system's own history could not have predicted, and ten per cent of the raw aggregate is forecastable from lagged real activity, so part of the shock is an endogenous response to the business cycle. We therefore call $v_1$ *the unanticipated component of monthly credit-market dislocation* throughout, and never an exogenous credit-supply shock. The estimator itself is untouched by the rejection: this is a rejection on the level of the aggregate, the predictable part is removed before the impact vector is formed, and the internal design remains unaffected by that predictability, and robust to non-invertibility should it be present (Plagborg-Møller and Wolf, 2021).

## 5.2 Impulse responses

**Table 13: Macroeconomic impulse responses**, per shock moving |EBP| by 25 basis points on impact. Point estimate is the Minnesota posterior mean; brackets are [16, 84] moving-block bootstrap percentiles on the same estimator. Log-level variables in percent; UNRATE in percentage points.

| Shock | Variable | $h = 6$ | $h = 12$ | $h = 24$ | $h = 48$ |
|---|---|---|---|---|---|
| $v_1$ credit | INDPRO | −1.48 [−1.52, −1.11] | −1.57 [−1.72, −1.09] | −1.11 [−1.49, −0.59] | −0.85 [−1.21, −0.44] |
| | UNRATE | +0.28 [+0.15, +0.42] | +0.28 [+0.18, +0.40] | +0.23 [+0.11, +0.41] | +0.11 [+0.03, +0.28] |
| | PCEPI | −0.42 [−0.50, −0.27] | −0.44 [−0.52, −0.27] | −0.45 [−0.57, −0.26] | −0.38 [−0.52, −0.19] |
| $v_2$ equity | INDPRO | −0.84 [−1.30, −0.33] | −1.40 [−2.03, −0.67] | −1.44 [−2.26, −0.57] | −1.16 [−1.90, −0.41] |
| | UNRATE | +0.10 [−0.09, +0.28] | +0.30 [+0.11, +0.50] | +0.45 [+0.19, +0.69] | +0.37 [+0.15, +0.60] |
| | PCEPI | −0.07 [−0.22, +0.06] | −0.19 [−0.43, −0.00] | −0.43 [−0.79, −0.11] | −0.47 [−0.88, −0.09] |
| $v_3$ volatility | INDPRO | −0.71 [−0.89, −0.28] | −0.79 [−1.16, −0.34] | −0.63 [−1.09, −0.21] | −0.51 [−0.84, +0.05] |
| | UNRATE | +0.36 [+0.11, +0.51] | +0.38 [+0.18, +0.53] | +0.31 [+0.18, +0.49] | +0.16 [+0.07, +0.33] |
| | PCEPI | −0.22 [−0.29, −0.11] | −0.22 [−0.30, −0.12] | −0.20 [−0.36, −0.08] | −0.07 [−0.30, +0.09] |

Every shock lowers industrial production, raises unemployment, depresses housing starts and house prices, and lowers consumer prices, with [16, 84] intervals excluding zero at the peak for every one of those responses. Nothing of the sort is imposed: the identification uses no sign restriction, and the sign convention fixes only the orientation of the axis and not the shape of the response.

Table 14 is where the central claim of this section can be checked. For each of the ten variables and each shock it gives the largest response over four years, the month at which it arrives, and the interval at that month. Twenty-seven of the thirty peaks have intervals excluding zero. Read down the columns the magnitudes are close: the peak output effects are −1.62%, −1.50% and −1.36%, and housing starts are the most shock-sensitive margin in the system under all

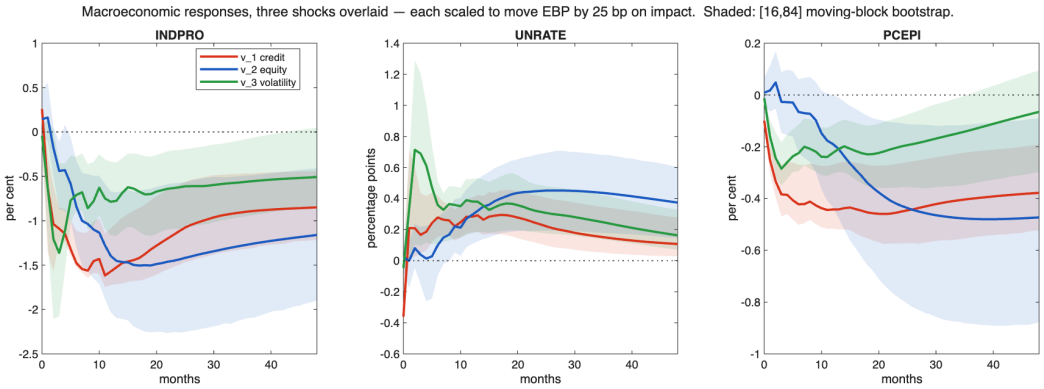



**Figure 5: The three shocks on common axes**, one panel per macro variable. Each is scaled to move the excess bond premium by 25 basis points on impact, so the vertical distance between the lines is a difference in effect at a common size; shaded areas are [16, 84] moving-block bootstrap bands.

**Table 14: Peak macroeconomic response and the horizon at which it occurs.** For each shock and variable, the entry is the response of largest absolute magnitude over $h = 0, \ldots, 48$, the month $h$ at which it is attained, and the [16, 84] moving-block bootstrap interval at that month. All responses are per shock moving the excess bond premium by 25 basis points on impact. INDPRO, HOUST, CSHPI, PCEPI and NRCREDIT are log levels and are reported in percent; UNRATE, FFR, RAI_BEX, UNC_BEX and EBP are in percentage points. An asterisk marks an interval that excludes zero. The financial block peaks at or near impact by construction, since that is where the shock lands; the one entry whose peak is the impact month and whose sustained response has the opposite sign is the unemployment response to the credit shock, discussed below.

| | $v_1$ credit | | | $v_2$ equity | | | $v_3$ volatility | | |
|---|---|---|---|---|---|---|---|---|---|
| Variable | Peak | $h$ | [16, 84] | Peak | $h$ | [16, 84] | Peak | $h$ | [16, 84] |
| INDPRO | −1.62* | 11 | [−1.74, −1.14] | −1.50* | 17 | [−2.22, −0.66] | −1.36* | 3 | [−2.07, −0.35] |
| UNRATE | −0.36* | 0 | [−0.41, −0.03] | +0.45* | 26 | [+0.19, +0.70] | +0.71* | 2 | [+0.10, +1.29] |
| HOUST | −5.18* | 6 | [−6.10, −2.47] | −6.36* | 11 | [−9.19, −2.48] | −6.69* | 2 | [−9.28, −3.47] |
| CSHPI | −0.56* | 6 | [−0.72, −0.32] | −3.22* | 41 | [−5.34, −0.79] | −0.90* | 6 | [−1.28, −0.51] |
| PCEPI | −0.46* | 21 | [−0.58, −0.27] | −0.48* | 39 | [−0.89, −0.11] | −0.29* | 3 | [−0.35, −0.18] |
| NRCREDIT | −0.64* | 23 | [−0.99, −0.21] | +0.33 | 11 | [−0.22, +0.78] | +0.52 | 48 | [−0.45, +0.72] |
| FFR | −0.09 | 2 | [−0.12, +0.01] | +0.30* | 9 | [+0.01, +0.54] | −0.25* | 3 | [−0.32, −0.14] |
| RAI_BEX | +0.46* | 0 | [+0.37, +0.55] | +0.45* | 0 | [+0.32, +0.73] | +0.89* | 0 | [+0.73, +1.07] |
| UNC_BEX | +0.27* | 0 | [+0.21, +0.36] | +0.28* | 5 | [+0.16, +0.38] | +0.45* | 1 | [+0.34, +0.56] |
| EBP | +0.27* | 2 | [+0.21, +0.32] | +0.37* | 1 | [+0.30, +0.45] | +0.35* | 1 | [+0.28, +0.39] |

three, at −5.18%, −6.36% and −6.69%. Read across the horizons they are not close at all, and that is the finding. The month at which the peak arrives ranges from two to forty-one, and it does so in a pattern that tracks the label of the shock rather than the variable. What separates the three is timing, not depth, and the ordering is economically legible.

- **The volatility shock is fastest.** Output troughs at three months, unemployment at two, housing starts at two, and the response then recovers: by $h = 48$ the output effect has halved and the price effect is no longer distinct from zero. This is the wait-and-see dynamic of Bloom (2009), in which real-options effects produce an abrupt pause in irreversible investment and hiring followed by a rebound once uncertainty resolves. That an identification using no timing restriction recovers this profile, and assigns it to the shock anchored on the VIX and to neither of the others, validates the labels. The profile is also consistent with the small but significant effects of daily-identified uncertainty shocks found by Alessandri et al. (2023) with the same bridge architecture, although, unlike Bloom (2009), the recovery shows no overshoot.

- **The credit shock is intermediate and persistent.** Output troughs at eleven months and prices at twenty-one, and both effects are still clearly negative at four years. Non-revolving consumer credit contracts 0.64% with a trough at twenty-three months, and this is the only one of the three to reduce the *quantity* of credit. Output down, prices down, credit quantity down and persistent is the textbook signature of an adverse credit-supply disturbance, reached here by a route entirely different from that of Gilchrist and Zakrajšek (2012), who identify the shock recursively, ordering the excess bond premium after the real variables, whereas the instrument here is a fourth-moment rotation of three daily prices.

- **The equity shock is slowest and works through wealth.** Output troughs at seventeen months and unemployment at twenty-six, the latest of the three, and the price level is still falling at $h = 48$. Its distinguishing feature is house prices, with a −3.22% peak at forty-one months, between three and six times what either of the other shocks does to the same variable. An adverse repricing of risk appetite transmits through the collateral and wealth channel, and it is the only one of the three that does not reduce the quantity of consumer credit.

**The impact month.** One entry of Table 14 is the peak only because it falls on impact, and its sign is the opposite of the sustained response. Unemployment falls 0.36 percentage points in the month of the credit shock, with an interval excluding zero, and is positive at every horizon from the first month onwards, reaching +0.29 at seventeen. Nothing in this design identifies a labour-market response within the calendar month in which a financial disturbance occurs, and the variance decomposition says the same from the other side, attributing 7.2% of the impact-month forecast error variance of unemployment to this shock against 1.3% of industrial production. We report the number and offer no interpretation of it; every statement about unemployment in this paper refers to the sustained response.

**The financial block.** Its peaks sit at impact or within two months of it for all three shocks, which is the architecture of the design and not a result: the shock lands on financial prices and the real block responds through the estimated dynamics of the system. What is unimposed is

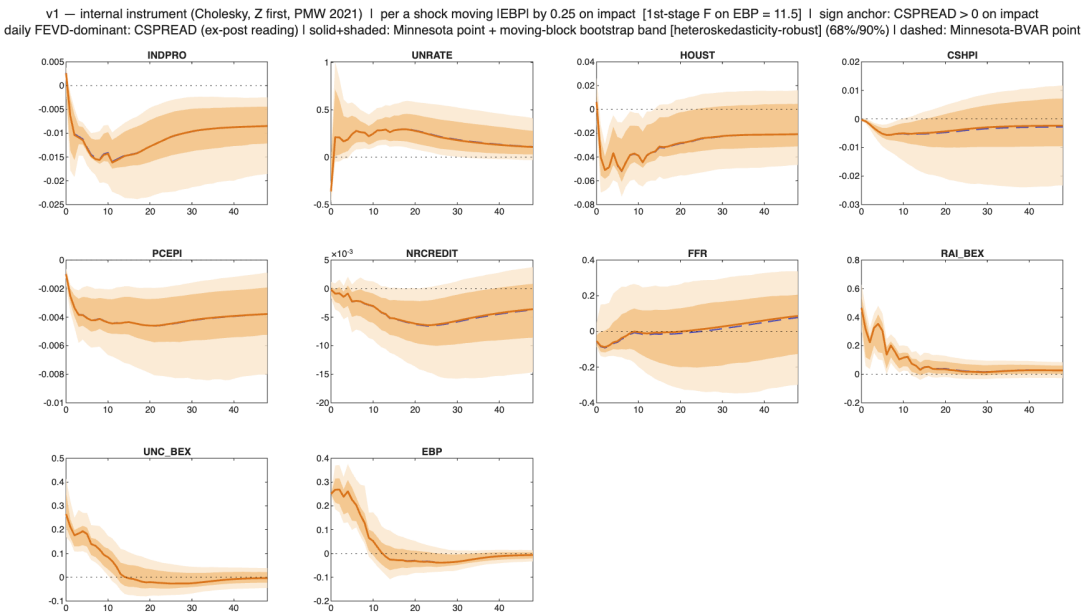



**Figure 6: Macroeconomic impulse responses to the credit shock** $v_1$, all ten variables. Solid line: Minnesota posterior mean; shaded: $[16, 84]$ and $[5, 95]$ moving-block bootstrap percentiles on the same estimator; dashed: Gaussian posterior band, for comparison. The responses to $v_2$ and $v_3$ are in Online Appendix F.9.

the ordering within the block, and it is coherent. The volatility shock produces by far the largest risk-aversion response, +0.89 against +0.46 and +0.45, and the largest uncertainty response; the equity shock has the most contained financial footprint, which is the arithmetic counterpart of its being the weakest of the three in the first stage.

The funds-rate responses are small relative to the earlier literature and are offered as a reading rather than a result. Two are distinguishable from zero: the volatility shock produces an easing of 0.25 percentage points at three months and the equity shock a *tightening* of 0.30 percentage points at nine. The asymmetry is economically coherent, with policy accommodating a spike in uncertainty and not leaning against an equity repricing whose inflation consequences arrive slowly, but it rests on responses of a few tenths of a percentage point. Re-estimating the whole system at $p_m \in \{12, 15, 18\}$ leaves the responses indistinguishable at $p_m = 18$, with correlations between 0.987 and 0.999 on every (shock, variable) pair, and changes them materially at $p_m = 12$, with correlations as low as 0.35: the specification sits on a plateau, and a shorter lag order truncates dynamics that are still building at the end of its own window (Online Appendix F.8). The equivalence with LP-IV holds in population up to horizon $p_m$; beyond fifteen months the responses are VAR extrapolations, whose bias–variance trade-off is documented by Li et al. (2024).

## 5.3 Variance decomposition

An impulse response states how a variable moves after a shock and not whether the shock matters. The decomposition is the object that separates the two, and it is also invariant to the normalisation. It is not a perfect object either: instrument noise attenuates every share by a

factor $\rho^2 \leq 1$ that differs across instruments, so the entries of Table 15 are lower bounds rather than point estimates (Plagborg-Møller and Wolf, 2022). The attenuation strengthens a large reported share rather than weakening it. Online Appendix C.10 gives the computation and the internal checks.

**Table 15: Share of the $h$-step forecast error variance attributable to the identified shock**, in percent, with [16, 84] posterior bands. Each column comes from its own augmented VAR, so shares *cannot* be summed across columns. The row for the instrument itself is an exact internal check: at $h = 0$ it must equal 100, and does. Entries without brackets are posterior medians for which the band was not retained by the estimation run.

| Shock | Variable | $h = 0$ | $h = 6$ | $h = 12$ | $h = 48$ |
|---|---|---|---|---|---|
| $v_1$ credit | INDPRO | 1.3 [0.4, 2.5] | 16.8 [11.8, 21.7] | **21.5** [14.1, 27.6] | 17.1 [7.6, 21.9] |
| | UNRATE | 7.2 [5.0, 10.0] | 4.6 [2.7, 7.7] | 6.6 [3.3, 11.5] | 8.3 [3.1, 13.5] |
| | PCEPI | 2.9 [1.6, 4.7] | 20.2 [14.4, 25.2] | **21.5** [14.1, 27.3] | 18.6 [8.3, 24.5] |
| | EBP | 16.2 [13.1, 19.7] | 24.6 | 21.3 | 18.2 |
| | RAI_BEX | 18.1 [15.1, 21.5] | 23.2 | 21.9 | 14.5 |
| $v_2$ equity | INDPRO | 0.0 | 1.0 [0.4, 3.0] | 2.9 [0.9, 6.1] | 5.7 [1.5, 10.3] |
| | UNRATE | 0.0 | 0.1 | 0.8 | 6.3 |
| | CSHPI | 1.0 | 7.9 [4.6, 11.4] | 9.1 [4.8, 13.3] | 6.9 |
| | EBP | 4.4 [2.5, 6.6] | 8.9 | 8.9 | 8.1 |
| $v_3$ volatility | INDPRO | 0.0 | 7.4 [4.4, 11.2] | 5.4 [2.6, 9.4] | 3.7 [1.4, 8.2] |
| | UNRATE | 0.1 | 11.1 [6.4, 15.9] | **11.7** [6.5, 17.1] | 10.7 [4.7, 16.5] |
| | RAI_BEX | **37.7** [34.2, 41.2] | 35.4 | 32.0 | 26.2 |
| | UNC_BEX | 22.8 [19.3, 26.4] | 27.7 | 24.9 | 20.9 |

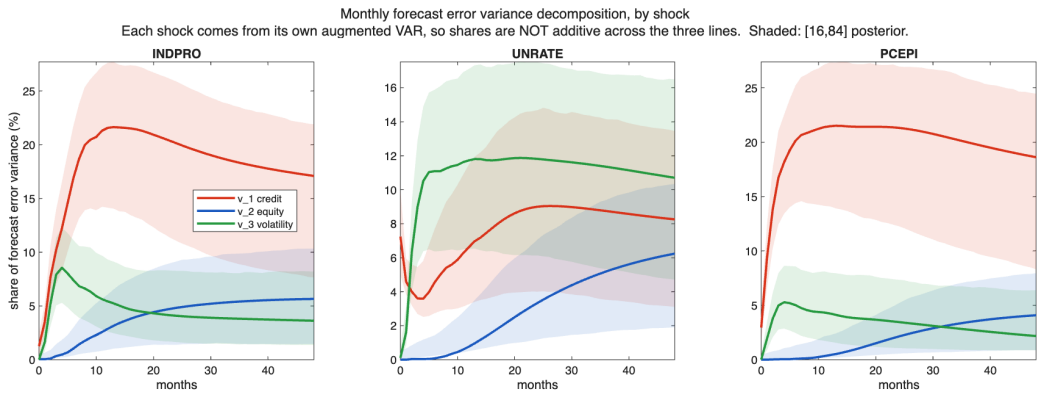



**Figure 7: Share of the $h$-step forecast error variance attributable to each identified shock**, with [16, 84] posterior bands. Each line comes from its own augmented VAR, so the three are *not* additive and do not sum to one. The rise of the credit share in industrial production between impact and one year is the transmission chain discussed below.

The decomposition first makes the transmission chain visible. At $h = 0$ the identified shock already accounts for between 4% and 38% of the forecast error variance of the financial variables, against 1.3% of industrial production; by $h = 12$ the industrial-production share of the credit shock has risen to 21.5%. The near-zero first stage against output in Table 12 and the large output response in Table 13 are two views of the same architecture.

It then settles the question an impulse response cannot. The credit shock accounts for 21.5% of

the twelve-month forecast error variance of industrial production and 21.5% of that of the price level, with bands excluding any small value. The response is therefore statistically distinguishable from zero and the shock producing it accounts for a fifth of the medium-run uncertainty about output, which is a stronger statement than the first on its own. The magnitude is not directly comparable with the risk-shock literature: Christiano et al. (2014) attribute close to 60 per cent of the variance of output growth to risk shocks in an estimated DSGE model, a different object from the twelve-month forecast-error share of industrial production reported here, which is moreover a lower bound. The ordering agrees with Caldara et al. (2016), who separate financial from uncertainty shocks by a penalty function and likewise find the financial one dominant. The dominance is not uniform. On unemployment the ordering reverses, with the volatility shock at 11.7% against the credit shock's 6.6%: credit owns output and prices, volatility owns the labour market. The table also confirms the labels by a route the identification never used, since $v_1$ accounts for 21.3% of the variance of the excess bond premium and $v_3$ for 37.7% of risk aversion and 24.9% of uncertainty, and no step in the identification was told what these series measure.

The equity shock is the instructive case, because it pairs a large response with a small share: its peak output effect of $-1.50\%$ is close to the largest in Table 14, yet it explains 2.9% of output variance at twelve months, about a seventh of the credit shock's share. The two statements are not in tension. The impulse response measures the effect per unit of shock under a normalisation that scales each shock to a common movement of EBP, whereas the decomposition measures the contribution of the shock as it actually occurs, so the equity shock has a large conditional effect and a small unconditional contribution. This is the clearest demonstration of why both objects have to be reported and why the comparison below uses the second.

## 5.4 Non-Gaussianity and macroeconomic importance

The identification isolates shocks by their non-Gaussianity and orders them by it; the macro stage measures how much of the economy's forecast uncertainty each accounts for. The two exercises share no data and no statistic, so it is worth asking whether they agree.

**Table 16: Daily non-Gaussianity against macroeconomic importance.** $\kappa_k$ is the realised excess kurtosis of the identified shock in the daily three-variable system; the FEVD share is from the monthly ten-variable augmented BVAR. Both are invariant to how the shock is scaled.

| Shock | $\kappa_k$ (daily) | Rank | FEVD INDPRO, $h = 12$ | $[16, 84]$ | Rank |
|---|---|---|---|---|---|
| Credit | 37.8 | 1 | 21.5% | [14.1, 27.6] | 1 |
| Equity | 12.4 | 2 | 2.9% | [ 0.9, 6.1] | 3 |
| Volatility | 8.6 | 3 | 5.4% | [ 2.6, 9.4] | 2 |

They agree on the shock that leads, and there is no arithmetic path by which one ordering could impose itself on the other: the kurtoses come from the fourth moments of three daily prices, the variance shares from a monthly BVAR in ten macroeconomic series, and the only object common to the two exercises is the identified shock series. Below the top the orderings cross, with the equity shock carrying the higher kurtosis and the smaller share and the volatility shock the reverse, which is a statement about loadings rather than about tails: the volatility shock has a far more concentrated loading structure, and a variance decomposition rewards concentration.

How much this establishes should be stated plainly. With three shocks, agreement on the leader would arise by chance one time in three, so it is a coherence check and not a test. The substantive content of this section is the transmission itself: three contractionary responses with bands excluding zero and no sign restriction imposed, three distinct and economically legible timing profiles, and a credit shock accounting for at least a fifth of medium-run output uncertainty.

# 6 Conclusions

This paper asks whether the disturbances underlying daily movements in equity prices, implied volatility and corporate credit spreads can be separated without imposing the contemporaneous sign, zero or timing restrictions that are commonly used for this purpose. The answer is affirmative, but the empirical evidence also clarifies what such an identification does and does not establish.

The first result is that the second and fourth moments provide different information about financial disturbances. The daily system is close to block-diagonal in variance, with credit, equity and volatility shocks each accounting for most of the variation in their respective market segment. The same decomposition looks very different in the tails. The credit shock is by far the most non-Gaussian component, despite accounting for only a small share of equity and volatility variance. This distinction is economically relevant because financial stress is concentrated in rare and large movements rather than in the typical variation of asset prices. The historical decompositions reinforce this interpretation: across major stress episodes, the credit and volatility shocks remain closely tied to the markets they are identified from, while the contribution of the equity shock to the market decline varies substantially across episodes.

The second result concerns the interpretation of the statistical rotation. The identification is performed without using macroeconomic variables, external proxies, crisis dates or economic sign restrictions. Nevertheless, several pieces of evidence external to the estimation point to the same interpretation. The shock with the largest fourth moment has the contemporaneous signature associated with credit-market stress and concentrates its largest realisations in the major credit dislocations of 2008 and 2020. The volatility component is instead concentrated in episodes characterised by sharp increases in implied uncertainty, and its correlation with external uncertainty measures is stronger than that of the other components. Most importantly, the leading credit direction is recovered from two non-overlapping crisis windows more than a decade apart. These exercises do not prove the economic interpretation of the shocks, but they make it difficult to attribute their labels to a particular event, sign restriction or feature of the full-sample estimation.

The third result is that the identified financial disturbances have different macroeconomic dynamics. All three shocks are contractionary for real activity and employment at their peak, but their persistence and transmission mechanisms differ. The volatility shock has a relatively short-lived effect, while the credit shock is more persistent and is the only shock that produces a sustained contraction in the quantity of credit. The equity shock operates more gradually and is particularly associated with the housing channel. At the twelve-month horizon, the credit shock

accounts for a substantial share of the forecast-error variance of output and prices, whereas the volatility shock is relatively more important for labour-market fluctuations. The evidence therefore does not support a single generic "financial shock": the timing and transmission of financial disturbances depend materially on which component of financial stress is affected.

A separate methodological result is important for interpreting these findings. The contraction used to map the fourth-order cumulant tensor into a matrix does not change the population eigendirections under the maintained independence assumptions, but it does change the associated eigenvalues and therefore the ordering of the identified shocks. In the data, the identity contraction ranks the shocks according to their realised excess kurtosis, whereas an all-ones contraction reverses the ranking of the two leading components. The choice of contraction is therefore not an innocuous computational detail when the empirical analysis subsequently compares shocks by their estimated non-Gaussianity. More generally, the result suggests that higher-moment identification should be accompanied by explicit invariance checks rather than by treating the contraction scheme as purely numerical.

The evidence also places clear limits on the interpretation of the results. The contemporaneous independence assumption is not exact, with the equity-volatility leverage relation providing the main exception, and under the block bootstrap the equity and volatility components are less sharply separated from each other than from the credit component. The monthly credit instrument remains partly predictable from lagged real activity, so it should be interpreted as an unanticipated component of credit-market conditions rather than as a clean exogenous credit-supply disturbance. In addition, the reported uncertainty bands do not propagate uncertainty from the daily rotation into the monthly BVAR, and the identification exploits non-Gaussianity but does not use the conditional heteroskedasticity of financial returns as an additional source of identification (Lewis, 2021); combining the two, and weak-identification-robust inference in the spirit of Drautzburg and Wright (2023), are natural extensions. These limitations are relevant to the strength, rather than the existence, of the empirical patterns reported here.

Taken together, the results suggest that identifying financial stress through the covariance structure alone can miss an economically important distinction between ordinary financial fluctuations and tail events. In this U.S. system, the component that dominates the tails is not the component that dominates typical financial variance, and that distinction survives several checks that do not enter the identification procedure itself. The macroeconomic evidence then shows that the three components have different persistence and transmission mechanisms rather than constituting interchangeable measures of financial tightening.

An important question remains open. The design uses only high-frequency financial prices for identification and a monthly macroeconomic panel for transmission, so it can in principle be applied to other countries and financial systems. Whether the prominence of the credit component reflects a particular feature of U.S. financial markets or a more general property of financial stress is therefore an empirical question for future work.

# Statement on the use of generative AI

In line with the University of Bologna guidelines, I declare the following use of generative AI tools (Claude, Anthropic: Sonnet 5; Opus 4.8, 4.9, 5.0, 5.1 and 5.5) between February and September 2026.

- *Code and computation.* Support in writing, debugging and documenting the MATLAB and Python code used for data preparation, estimation and bootstrap inference (Sections 3–5 and the Online Appendix), and in running computationally intensive auxiliary checks. All code was run and checked by the author, and every number reported comes from the author's own estimation pipeline.
- *Study.* Explanations of the econometric methods used in the thesis (identification from higher moments, internal instruments, bootstrap inference), always checked against the original sources cited.
- *Language.* Revision of the English of text drafted by the author, in all sections and in the Online Appendix.

The research question, the identification strategy, the choice of methods, the interpretation of the results and the conclusions are the author's own, and the author takes full responsibility for the content of the thesis.

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